Presentation for the “Rough paths” exam

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Outline

Preliminaries

Stochastic sewing lemma in spaces of mixed integrability

Rough stochastic analysis

Rough stochastic filtering

Connection rough and stochastic filtering

Preliminaries

Stochastic sewing lemma in spaces of mixed integrability

Random variables of mixed integrability

Let $m \geq 1$ be a real number and fix a sub-σ-field $\mathcal{F} \subset \mathcal{G}$. For an m-integrable random variable X with values in a Banach space $(E, |\cdot|_E)$, we define the conditional L_m-norm as the random variable

$$\big\|X \mid \mathcal{F}\big\|_m := \big[\mathbb{E}\big(|X|_E^m \mid \mathcal{F}\big)\big]^{1/m}.$$

For an extended real number $n \in [1, \infty]$, we then define a space $L_{m,n}$ consisting of all $X \in L_{m\wedge n}$ such that

$$\|X\|_{m,n} := \big\|\big\|X \mid \mathcal{F}\big\|_m\big\|_n < \infty.$$

Basic properties: for any $1 \leq m \leq n \leq +\infty$ (with $n < +\infty$):

- $(L_{m,n}, ||\cdot||_{m,n})$ is a Banach space which coincides with L_m whenever $m = n < +\infty$;
- the following continuous embeddings hold:

$$L_n \hookrightarrow L_{m,n} \hookrightarrow L_m.$$

Theorem 1 ([FHL24], Theorem 2.8)

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ be a filtered probability space and let $2 \leq m \leq n \leq \infty$ be fixed, $m < \infty$. Let $A = (A_{s,t})_{(s,t)\in\Delta}$ be a stochastic process in a finite Banach space W such that $A_{s,s} = 0$ and $A_{s,t}$ is $\mathcal{F}_t$-measurable for every $(s, t) \in \Delta$.

(i) Suppose that there are finite constants $\Gamma_1, \Gamma_2 \geq 0$ and $\varepsilon_1, \varepsilon_2 > 0$ such that for any $0 \leq s \leq u \leq t \leq T$,

$$\|\mathbb{E}_s[\delta A_{s,u,t}]\|_n \leq \Gamma_1(t-s)^{1+\varepsilon_1}, \quad \Big\|\big\|\delta A_{s,u,t} \mid \mathcal{F}_s\big\|_m\Big\|_n \leq \Gamma_2(t-s)^{\frac{1}{2}+\varepsilon_2}.$$

Theorem 1 (continued)

Then, there exists a unique stochastic process (up to modifications) γ with values in W satisfying the following properties:

- $\gamma_0 = 0$, γ *is* $\{\mathcal{F}_t\}$*-adapted,* γ_t *is* L_m*-integrable for each* $t \in [0, T]$*;*
- *there are positive constants* $C_1 = C_1(\varepsilon_1)$, $C_2 = C_2(\varepsilon_2)$ *such that for every* $(s, t) \in \Delta$*,*

$$\big\|\big\|\|\gamma_t - \gamma_s - A_{s,t} \mid \mathcal{F}_s\|\big\|_m\big\|_n \leq C_1\Gamma_1(t-s)^{1+\varepsilon_1} + C_2\Gamma_2(t-s)^{\frac{1}{2}+\varepsilon_2},$$

and

$$\|\mathbb{E}_s(\gamma_t - \gamma_s - A_{s,t})\|_n \leq C_1\,\Gamma_1(t-s)^{1+\varepsilon_1}.$$

Theorem 1 (continued)

(ii) Suppose furthermore that for each $s \in [0, T]$, the map $t \mapsto A_{s,t}$ is a.s. càdlàg (resp. continuous) on $[s, T]$, and there are finite constants $\varepsilon_3 > 0$ and $\Gamma_3 \geq 0$ such that for any $(s, t) \in \Delta$,

$$\sup_{u \in [(s+t)/2, t]} \| \| \delta A_{s,(s+t)/2,u} \mid \mathcal{F}_s \|_m \|_n \leq \Gamma_3 (t-s)^{\frac{1}{m}+\varepsilon_3}.$$

Let $\mathcal{P} = \{0 = t_0 < t_1 < \cdots < t_N = T\}$ be a partition of $[0, T]$ and define for each $t \in [0, T]$,

$$A_t^{\mathcal{P}} := \sum_{t_i \leq s < t_{i+1} \leq t} A_{t_i, t_{i+1}}.$$

Then γ has a càdlàg (resp. continuous) version, denoted by the same notation, and for this version, we have

$$\Big\| \| \sup_{t \in [0,T]} \mid A_t^{\mathcal{P}} - \gamma_t \mid \big| \mathcal{F}_0 \|_m \Big\|_n \leq C |\mathcal{P}|^{\varepsilon_1 \wedge \varepsilon_2 \wedge \varepsilon_3} (\Gamma_1 + \Gamma_2 + \Gamma_3),$$

where $|\mathcal{P}| := \sup_i |t_{i+1} - t_i|$ and C is some constant depending on $T, m, \varepsilon_1, \varepsilon_2, \varepsilon_3$.

Preliminaries

Rough stochastic analysis

Notation for Rough Paths

- For a fixed $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, a rough path is denoted by

$$\mathbf{Y} = (Y, \mathbb{Y}) \in \mathcal{C}^{\alpha}([0, T], \mathbb{R}^{d_Y})$$

  and we endow this space with the pseudo-distance:

$$\rho_{\alpha}(\mathbf{Y}, \overline{\mathbf{Y}}) = |Y - \overline{Y}|_{\alpha} + |\mathbb{Y} - \overline{\mathbb{Y}}|_{2\alpha},$$

- The inhomogeneous seminorm is

$$|\mathbf{Y}|_{\alpha} = |Y|_{\alpha} + |\mathbb{Y}|_{2\alpha},$$

  and the homogeneous seminorm is

$$\|\mathbf{Y}\|_{\alpha} = |Y|_{\alpha} \vee \sqrt{|\mathbb{Y}|_{2\alpha}}.$$

- The bracket of a rough path $\mathbf{Y} = (Y, \mathbb{Y})$ is given by the implicit definition

$$(\delta Y_{s,t}) \otimes (\delta Y_{s,t}) = \mathbb{Y}_{s,t} + \mathbb{Y}^{\top}_{s,t} + \delta[\mathbf{Y}]_{s,t}.$$

Rough stochastic integration

We wish to give a precise meaning to the integral

$$\int_0^T (f_t, f_t{}')(X_t, \omega; \mathbf{Y})\, d\mathbf{Y}_t. \tag{1}$$

To this end, we must carefully specify the function spaces in which the coefficients $(f_t, f_t{}')$ take values.

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If we momentarily ignore the dependence on X_t, then we need a notion of a *stochastic controlled rough path*.

Stochastic controlled rough path

Let $p \in [2, \infty)$ and $q \in [p, \infty]$. We say that $(F, F') = (F_t, {F_t}')(\omega, Y)$ is a stochastic controlled rough path in $\mathbf{D}_Y^{2\alpha} L_{p,q}\big([0, T], \Omega; \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X})\big)$ if

$$(F, F') : \Omega \times [0, T] \to \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X}) \times \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X}))$$

are progressively measurable and

$$\|(F, F')\|_{Y;2\alpha;p,q} := \sup_{0 \le s < t \le T} \Bigg(\frac{\|\mathbb{E}\big(|\delta F_{s,t}|^p \,\big|\, \mathcal{F}_s\big)^{1/p}\|_q}{|t-s|^\alpha} + \|F'_t\|_q$$

$$+ \frac{\|\mathbb{E}\big(|\delta {F'}_{s,t}|^p \,\big|\, \mathcal{F}_s\big)^{1/p}\|_q}{|t-s|^\alpha} + \frac{\|\mathbb{E}\big(\,\delta F_{s,t} - F'_s\,\delta Y_{s,t} \,\big|\, \mathcal{F}_s\big)\|_q}{|t-s|^{2\alpha}} \Bigg) < +\infty.$$

Let $p \in [2, \infty)$ and $q \in [p, \infty]$. We say that $(F, F') = (F_t, F_t{}')(\omega, Y)$ is a stochastic controlled rough path in $\mathbf{D}_Y^{2\alpha} L_{p,q}\big([0, T], \Omega; \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X})\big)$ if

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are progressively measurable and

$$\|(F, F')\|_{Y;2\alpha;p,q} := \sup_{0 \le s < t \le T} \left(\frac{\|\mathbb{E}\big(|\delta F_{s,t}|^p \,\big|\, \mathcal{F}_s\big)^{1/p}\|_q}{|t-s|^\alpha} + \|F'_t\|_q \right.$$

$$\left. + \frac{\|\mathbb{E}\big(|\delta F'_{s,t}|^p \,\big|\, \mathcal{F}_s\big)^{1/p}\|_q}{|t-s|^\alpha} + \frac{\|\mathbb{E}\big(\delta F_{s,t} - F'_s\,\delta Y_{s,t} \,\big|\, \mathcal{F}_s\big)\|_q}{|t-s|^{2\alpha}} \right) < +\infty.$$

If (F, F') are deterministic, we write

$$(F, F') \in \mathcal{D}_Y^{2\alpha}([0, T], \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X}))$$

and we retrieve the classical notion of controlled rough paths (Definition 4.6 in [FH20]).

Definition of the rough stochastic integral

We can give meaning to

$$\int_0^T (f_t, f_t{}')(X_t, \omega; \mathbf{Y})\, d\mathbf{Y}_t \qquad (1)$$

using the previous definition with

$$(F_t, F_t')(\omega, \mathbf{Y}) = \big(f_t(X_t, \omega; \mathbf{Y}),\ ((Df_t)f_t + f_t')(X_t, \omega; \mathbf{Y})\big).$$

Infact, it holds:

Theorem 2 ([FHL24], Theorem 4.2)

If $q = \infty$ or f is linear in x and $q = p$, we obtain an $L_{p,q}$-integrable solution $X = X_t^{\mathbf{Y}}(\omega)$ to (1) in the sense that it is continuous and adapted, and the following are satisfied:

- $(f_t(X_t, \omega; \mathbf{Y}),\ ((Df_t)f_t + f_t')(X_t, \omega; \mathbf{Y})) \in \mathbf{D}_{\mathbf{Y}}^{2\alpha} L_{p,q}\big([0, T], \Omega; \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X})\big)$;
- $\delta X_{s,t} = f_s(X_s, \omega; \mathbf{Y})\, \delta Y_{s,t} + \big((Df_s)f_s + f_s'\big)(X_s, \omega; \mathbf{Y})\, \mathbb{Y}_{s,t} + X_{s,t}^{\natural}$ *with*

$$\sup_{0 \le s < t \le T} \frac{\|\mathbb{E}(|X_{s,t}^{\natural}|^p \mid \mathcal{F}_s)^{1/p}\|_q}{|t-s|^{2\alpha}} < +\infty \qquad \textit{and} \qquad \sup_{0 \le s < t \le T} \frac{\|\mathbb{E}(X_{s,t}^{\natural} \mid \mathcal{F}_s)\|_q}{|t-s|^{3\alpha}} < +\infty.$$

Stochastic controlled vector field

Let $p \in [2, \infty)$, $q \in [p, \infty]$, $\beta = N + \kappa > 1$ with $N \in \mathbb{N}_{\geq 0}$ and $\kappa \in (0, 1]$. A pair

$$(f, f') = (f_t(x, \omega; \mathbf{Y}),\ f'_t(x, \omega; \mathbf{Y}))$$

is a stochastic controlled vector field in $\mathbf{D}_Y^{2\alpha} \mathcal{C}_b^{\beta} L_{p,q}([0,\ T], \Omega; \mathbb{R}^{d_X})$:

(1) $(f, f') : \Omega \times [0,\ T] \longrightarrow \mathcal{C}_b^{\beta}(\mathbb{R}^{d_X}; \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X})) \times \mathcal{C}_b^{\beta-1}(\mathbb{R}^{d_X}; \mathrm{Lin}(\mathbb{R}^{d_Y},\ \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X}))$ is progressively measurable, and

$$\|(f, f')\|_{\beta} \ := \ \sup_{t \in [0,\ T]} \|\ |\ f_t\ |_{C_b^{\beta}}\ \|_q \ + \ \sup_{t \in [0,\ T]} \|\ |\ f'_t\ |_{C_b^{\beta-1}}\ \|_q \ < \ \infty.$$

(2) For each $j = 0, \ldots, N$,

$$(D_x^j f,\ D_x^j f') \in \mathbf{D}_Y^{2\alpha} L_{p,q}([0,\ T], \Omega; \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X})),$$

Stochastic controlled vector field

We endow $\mathbf{D}_Y^{2\alpha}\mathcal{C}_b^{\beta}L_{p,q}([0,T],\Omega;\mathbb{R}^{d_X})$ with the seminorm

$$[\![f,f']\!]_{Y;2\alpha;\beta;p,q} := \sum_{j=0}^{N}[\![\delta D_x^j f]\!]_{\alpha;p,q} + \sum_{j=0}^{N-1}\left([\![\delta D_x^j f']\!]_{\alpha;p,q} + [\![R^{D_x^j f}]\!]_{2\alpha;p,q}\right) < \infty.$$

In the above,

$$D_x^0 f = f, \qquad \delta f_{s,t} := f_t - f_s, \qquad R^f_{s,t} := f_t - f_s - f'_s\,\delta Y_{s,t},$$

and for any $\zeta = \zeta_{s,t}(x,\omega)$ we set

$$[\![\zeta]\!]_{\gamma;p,q} := \sup_{0\le s<t\le T}\frac{\left\|\mathbb{E}\left(\sup_{x\in\mathbb{R}^{d_X}}|\zeta_{s,t}(x,\omega)|^p \,\middle|\, \mathcal{F}_s\right)^{1/p}\right\|_q}{|t-s|^\gamma}.$$

If f, f' and ζ are deterministic, we simply write

$$(f,f') \in \mathcal{D}_Y^{2\alpha}C_b^{\beta}([0,T];\mathbb{R}^{d_X}), \qquad [\![f,f']\!]_{Y;2\alpha;\beta}, \qquad \text{and} \qquad [\![\zeta]\!]_{\gamma}.$$

Outline

Preliminaries

Rough stochastic filtering

- Dynamics of signal and observation
- Rough Kallianpur-Striebel and robustness
- Rough Zakai equation in spaces of measures: existence
- Rough Zakai equation in spaces of measures: uniqueness
- Rough Kushner-Stratonovich equation

Connection rough and stochastic filtering

Rough stochastic filtering

Dynamics of signal and observation

From the original model to the simplified dynamics (non–degenerate case)

Stochastic dynamics under the original measure

$$\text{(S)} \quad dX_t = \bar{b}(t, X_t, Y_t)\, dt + \sigma(t, X_t, Y_t)\, dB_t + \bar{f}(t, X_t, Y_t)\, dB_t^{\perp},$$

$$\text{(O)} \quad dY_t = \bar{h}(t, X_t, Y_t)\, dt + k(t, Y_t)\, dB_t^{\perp}.$$

- $(\Omega, \mathcal{F}, (\mathcal{F})_{t\in[0,T]}, \mathbb{P}^o)$ a filtered probability space, equipped with complete and indipendent sub-filtrations $(\mathcal{F}^B)_{t\in[0,T]}$, $(\mathcal{F}^{B^\perp})_{t\in[0,T]}$, which supports independent Brownian motions B and $B^\perp$of repsective dimension d_B and d_W;
- X of dimension d_X and Y of dimension $d_Y = d_W$;
- $k(t, y)$ is invertible for all (t, y);
- $\bar{h}$ admits the decomposition

$$\bar{h}(t, x, y) = h_1(t, y) + k(t, y)\, h_2(t, x, y),$$

allowing to absorbe h_2 in

$$W_t := \int_0^t h_2(r, X_r Y_r)\, dr + B_t^{\perp}.$$

Change of measure

Assumtion CP-M: For every $t \in [0, T]$ we have

$$\mathbb{E}[\widetilde{Z}_t] = 1, \qquad \text{where} \qquad \widetilde{Z}_t := \exp\left(- \int_0^t h_2(r, X_r, Y_r)^\top dB_r^\perp - \frac{1}{2} \int_0^t |h_2(r, X_r, Y_r)|^2 \, dr \right).$$

By a Girsanov change of measure, W then becomes a new Brownian motion, independent of B under the probability $\mathbb{P}$ given by the density

$$\left. \frac{d\mathbb{P}}{d\mathbb{P}^0} \right|_{\mathcal{F}_T} = \widetilde{Z}_T,$$

the inverse of which is given by

$$Z_T := \exp(I_T) := \exp\left(\int_0^T h_2(r, X_r, Y_r)^\top \, dW_r - \frac{1}{2} \int_0^T |h_2(r, X_r, Y_r)|^2 \, dr \right).$$

Stochastic dynamics

Under $\mathbb{P}$:

$$\text{(S)} \quad dX_t = b(t, X_t, Y_t)\, dt + \sigma(t, X_t, Y_t)\, dB_t + f(t, X_t, Y_t)\, dY_t$$

$$\text{(O)} \quad dY_t = k(t, Y_t)\, dW_t, \qquad Y_0 = 0$$

$$\text{(E)} \quad dZ_t = Z_t\, h(t, X_t, Y_t)\, dY_t, \qquad Z_0 = 1$$

where:

- the dimensions are d_X and $d_Y = d_W$;
- k invertible;
- $\mathbb{P}$-Brownian motions $W \perp B$ and $Y \perp B$;
- Y is a $\mathbb{P}$-local martingale with covariation dynamics:

$$d\langle Y, Y\rangle_t = k(t, Y_t)\, k(t, Y_t)^\top\, dt.$$

For $\alpha \in (1/3,\, 1/2]$, we fix a rough path

$$\mathbf{Y} \in \mathcal{C}^{0,\alpha,1} = \left\{ \mathbf{Y} \in \mathcal{C}^{0,\alpha}([0,\, T])^{1} \;\middle|\; t \mapsto \delta[\mathbf{Y}]_{0,t} \in C^{1}([0,\, T]) \right\}$$

and we consider the following RSDEs:

$$\text{(S)} \quad dX_t^{\mathbf{Y}} = b(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, dt + \sigma(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, dB_t + f(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, d\mathbf{Y}_t, \tag{2}$$

$$\text{(E)} \quad dZ_t^{\mathbf{Y}} = Z_t^{\mathbf{Y}}\, h(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, d\mathbf{Y}_t, \qquad Z_0^{\mathbf{Y}} = 1 \in \mathbb{R}. \tag{3}$$

[1] $\mathcal{C}^{0,\alpha}$ is the Polish space of $\alpha-$Hölder rough paths obtained as $\rho_\alpha-$ closure of smooth rough paths (genuine rough path with the additional property that the V-valued (resp. $V \otimes V$-valued) maps $X_\bullet$ and $\mathbb{X}_{s,\bullet}$ are smooth, for every basepoint s.)

Rough dynamics

For $\alpha \in (1/3,\, 1/2]$, we fix a rough path

$$\mathbf{Y} \in \mathcal{C}^{0,\alpha,1} = \left\{\, \mathbf{Y} \in \mathcal{C}^{0,\alpha}([0,\, T])^{\,1} \;\Big|\; \mid t \mapsto \delta[\mathbf{Y}]_{0,t} \in C^1([0,\, T]) \,\right\}$$

and we consider the following RSDEs:

$$\text{(S)} \quad dX_t^{\mathbf{Y}} = b(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, dt + \sigma(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, dB_t + f(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, d\mathbf{Y}_t, \tag{2}$$

$$\text{(E)} \quad dZ_t^{\mathbf{Y}} = Z_t^{\mathbf{Y}}\, h(t, X_t^{\mathbf{Y}}; \mathbf{Y})\, d\mathbf{Y}_t, \qquad Z_0^{\mathbf{Y}} = 1 \in \mathbb{R}. \tag{3}$$

First question: whether and when the equations above admit a solution, and in what sense.

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[1] $\mathcal{C}^{0,\alpha}$ is the Polish space of $\alpha-$Hölder rough paths obtained as $\rho_\alpha-$ closure of smooth rough paths (genuine rough path with the additional property that the V-valued (resp. $V \otimes V$-valued) maps $X_\bullet$ and $\mathbb{X}_{s,\bullet}$ are smooth, for every basepoint s.)

Assumption E – Regularity of Coefficients

For any fixed rough path $\mathbf{Y} \in \mathcal{C}^{0,\alpha,1}$, the measurable functions

$$b(\cdot,\,\cdot;\mathbf{Y})\;:[0,\,T]\times\mathbb{R}^{d_X}\to\mathbb{R}^{d_X}, \qquad \sigma(\cdot,\,\cdot;\mathbf{Y}):[0,\,T]\times\mathbb{R}^{d_X}\to\mathrm{Lin}(\mathbb{R}^{d_B},\mathbb{R}^{d_X})\equiv\mathbb{R}^{d_X\times d_B},$$

$$f(\cdot,\,\cdot;\mathbf{Y}):[0,\,T]\times\mathbb{R}^{d_X}\to\mathrm{Lin}(\mathbb{R}^{d_Y},\mathbb{R}^{d_X})\equiv\mathbb{R}^{d_X\times d_Y}, \quad h(\cdot,\,\cdot;\mathbf{Y}):[0,\,T]\times\mathbb{R}^{d_X}\to\mathrm{Lin}(\mathbb{R}^{d_Y},\mathbb{R})\equiv\mathbb{R}^{1\times d_Y}$$

satisfy:

(1) **Deterministic bounded Lipschitz vector fields:** the map $t\mapsto b(t,\cdot;\mathbf{Y})\in\mathcal{C}_b^1(\mathbb{R}^{d_X};\,\mathbb{R}^{d_X})$ is strongly measurable and globally bounded, and there is a constant $\|b(\cdot,\cdot;\mathbf{Y})\|_{\mathrm{Lip}}>0$ such that

$$\sup_{t\in[0,T]}\;\sup_{\substack{x,y\in\mathbb{R}^{d_X}\\ x\neq y}}\frac{|b(t,x;\mathbf{Y})-b(t,y;\mathbf{Y})|}{|x-y|}\;\leq\;\|b(\cdot,\cdot;\mathbf{Y})\|_{\mathrm{Lip}}\;<\;+\infty,$$

with similar condition on σ.

(2) **Controlled regularity of f:** there exists a function $f'(\cdot,\cdot;\mathbf{Y}) : [0,T] \times \mathbb{R}^{d_X} \to \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}^{d_X})) \equiv \mathrm{Lin}(\mathbb{R}^{d_Y} \otimes \mathbb{R}^{d_Y}, \mathbb{R}^{d_X})$ and a constant $\beta \in \left(\frac{1}{\alpha}, 3\right]$ such that

$$[t \mapsto (f(t,\cdot;\mathbf{Y}), f'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_Y^{2\alpha}\mathcal{C}_b^{\beta},$$

with $Y = \pi_1\mathbf{Y}$.

(3) **Controlled regularity of h:** there exists a function $h'(\cdot,\cdot;\mathbf{Y}) : [0,T] \times \mathbb{R}^{d_X} \to \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathrm{Lin}(\mathbb{R}^{d_Y}, \mathbb{R}))$ such that

$$[t \mapsto (h(t,\cdot;\mathbf{Y}), h'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_Y^{2\alpha}\mathcal{C}_b^{2}.$$

with $Y = \pi_1\mathbf{Y}$.

(4) **Initial condition:** assume $X_0^{\mathbf{Y}} \in L^0(\Omega, \mathcal{F}_0; \mathbb{R}^{d_X})$.

Theorem 3 ([FHL24], Theorem 2.6)

Under Assumption E, there exists a unique $L_{2,\infty}$-integrable solution $X^{\mathbf{Y}}$ to Equation (2) in the sense that is the unique continuous and adapted process such that

$$X_t^{\mathbf{Y}} - X_s^{\mathbf{Y}} = \int_s^t b(r, X_r^{\mathbf{Y}}; \mathbf{Y})\, dr + \sum_{\theta=1}^{d_B} \int_s^t \sigma_\theta(r, X_r^{\mathbf{Y}}; \mathbf{Y})\, dB_r^\theta + \sum_{\kappa=1}^{d_Y} f_\kappa(s, X_s^{\mathbf{Y}}; \mathbf{Y})\, \delta Y_{s,t}^\kappa$$
$$+ \sum_{\kappa,\lambda=1}^{d_Y} \left(\sum_{j=1}^{d_X} \partial_{x_j} f_\lambda(s, X_s^{\mathbf{Y}}; \mathbf{Y}) f_j^\kappa(s, X_s^{\mathbf{Y}}; \mathbf{Y}) + f'_{\lambda\kappa}(s, X_s^{\mathbf{Y}}; \mathbf{Y}) \right) \mathbb{Y}_{s,t}^{\kappa\lambda} + X_{s,t}^\natural,$$

$$\sup_{0 \le s < t \le T} \frac{\left\| \mathbb{E}(|X_{s,t}^\natural|^2 \mid \mathcal{F}_s^B)^{1/2} \right\|_\infty}{|t-s|^{2\alpha}} < +\infty, \qquad \sup_{0 \le s < t \le T} \frac{\left\| \mathbb{E}(|X_{s,t}^\natural| \mid \mathcal{F}_s^B) \right\|_\infty}{|t-s|^{3\alpha}} < +\infty.$$

Theorem 4 ([BCN24], Theorem 3.11)

Under Assumption E, there exists a unique L_2-integrable solution $Z^{\mathbf{Y}}$ to Equation (3) in the sense that is the unique process such that

$$Z_t^{\mathbf{Y}} - Z_s^{\mathbf{Y}} = \sum_{\kappa=1}^{d_Y} Z_s^{\mathbf{Y}}\, h_\kappa(s, X_s^{\mathbf{Y}}; \mathbf{Y})\, \delta Y_{s,t}^{\kappa} + \sum_{\kappa,\lambda=1}^{d_Y} \Big(Z_s^{\mathbf{Y}}\, h_\lambda(s, X_s^{\mathbf{Y}}; \mathbf{Y}) h_\kappa(s, X_s^{\mathbf{Y}}; \mathbf{Y}) + \sum_{j=1}^{d_X} \partial_{x_j} h_\lambda(s, X_s^{\mathbf{Y}}; \mathbf{Y}) Z_s^{\mathbf{Y}}\, f_j^{\kappa}(s, X_s^{\mathbf{Y}}; \mathbf{Y}) + Z_s^{\mathbf{Y}}\, h'_{\lambda\kappa}(s, X_s^{\mathbf{Y}}; \mathbf{Y}) \Big) \mathbb{Y}_{s,t}^{\kappa\lambda} + Z_{s,t}^{\natural},$$

$$\sup_{0\le s<t\le T} \frac{\left\| Z_{s,t}^{\natural} \right\|_2}{|t-s|^{2\alpha}} < +\infty, \qquad \sup_{0\le s<t\le T} \frac{\left\| \mathbb{E}(|Z_{s,t}^{\natural}| \mid \mathcal{F}_s^B) \right\|_2}{|t-s|^{3\alpha}} < +\infty.$$

Define, for $t \in [0, T]$,

$$I_t^{\mathbf{Y}} = \int_0^t h(r, X_r^{\mathbf{Y}}; \mathbf{Y})\, d\mathbf{Y}_r - \frac{1}{2} \int_0^t h(r, X_r^{\mathbf{Y}}; \mathbf{Y})\, [\dot{\mathbf{Y}}]_r\, h(r, X_r^{\mathbf{Y}}; \mathbf{Y})^\top\, dr.$$

Lemma 5 ([BFS25], Proposition 4.14)

Under Assumption E, $I_t^{\mathbf{Y}}$ is exponentially integrable uniformly in t, i.e.

$$\sup_{t \in [0,T]} \mathbb{E}\left(e^{I_t^{\mathbf{Y}}}\right) < \infty,$$

and moreover

$$e^{I_t^{\mathbf{Y}}} = Z_t^{\mathbf{Y}}$$

where $Z_t^{\mathbf{Y}}$ is the solution given by Theorem 4.

Rough stochastic filtering

Rough Kallianpur-Striebel and robustness

The stochastic Kallianpur–Striebel Theorem

Recall we want to compute, for any $\varphi \in b\mathcal{B}(\mathbb{R}^{d_X})$,

$$\varsigma_t(\varphi) = \mathbb{E}^\circ\left[\varphi(X_t) \mid \mathcal{F}_t^Y\right].$$

where $(\mathcal{F}_t^Y)$ is the observation filtration.

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Theorem 6 ([CP24], Theorem 3.5)

Granted Assumption CP-M, for every $\varphi \in b\mathcal{B}(\mathbb{R}^d)$ and $t \in [0, T]$,

$$\varsigma_t(\varphi) = \frac{\mathbb{E}[Z_t \varphi(X_t) \mid \mathcal{F}_t^Y]}{\mathbb{E}[Z_t \mid \mathcal{F}_t^Y]} =: \frac{\mu_t(\varphi)}{\mu_t(1)} \quad \mathbb{P}(\mathbb{P}^0)\text{-a.s.}$$

where $\mathcal{F}_t^Y$ is the observation filtration and $\mu = \{\mu_t, t \geq 0\}$ is the so-called unnormalised filter.

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where $\mathcal{F}_t^Y$ is the observation filtration and $\mu = \{\mu_t, t \geq 0\}$ is the so-called unnormalised filter.

Problem: When does there exist a continuous functional Φ on the path space such that

$$\mathbb{E}^\circ\left[\varphi(X_t) \mid \mathcal{F}_t^Y\right] = \Phi(Y)?$$

In view of the stochastic unnormalised filter

$$\mu_t(\omega)(\varphi) := \mathbb{E}\big(\varphi(X_t)Z_t \mid \mathcal{F}_t^Y\big)(\omega)$$

we define the **rough unnormalised filter**

$$\mu_t^{\mathbf{Y}}(\varphi) := \mathbb{E}\big(\varphi(X_t^{\mathbf{Y}})Z_t^{\mathbf{Y}}\big), \qquad t \in [0, T].$$

Notice that it is well-defined since $X_t^{\mathbf{Y}} \in L_{2,\infty}$ and $Z_t^{\mathbf{Y}} \in L^2$.

Rough Kallianpur-Striebel

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Notice that it is well-defined since $X_t^{\mathbf{Y}} \in L_{2,\infty}$ and $Z_t^{\mathbf{Y}} \in L^2$.

We call

$$\mathbf{Y} \mapsto \varsigma_t^{\mathbf{Y}}(\varphi) := \frac{\mu_t^{\mathbf{Y}}(\varphi)}{\mu_t^{\mathbf{Y}}(1)}$$

the **rough Kallianpur–Striebel map**.

“Distance” between stochastic controlled vector fields

Given $Y, \bar{Y} \in C^{\alpha}([0,T];\mathbb{R}^{d_Y})$ and $(g,g') \in \mathcal{D}_Y^{2\alpha} C_b^{\beta}$, $(\bar{g},\bar{g}') \in \mathcal{D}_{\bar{Y}}^{2\alpha} C_b^{\beta}$ $\beta = N+\kappa > 1$, we denote by

$$\llbracket g, g'; \bar{g}, \bar{g}' \rrbracket_{Y,\bar{Y};2\alpha;\beta} := \sum_{i=0}^{N} \llbracket \delta D_x^i (g-\bar{g}) \rrbracket_{\alpha} + \sum_{j=0}^{N-1} \left(\llbracket \delta D_x^j (g'-\bar{g}') \rrbracket_{\alpha} + \llbracket R^{D_x^j g} - \bar{R}^{D_x^j \bar{g}} \rrbracket_{2\alpha} \right),$$

with

$$\bar{R}_{s,t}^{D_x^j \bar{g}}(x) := D_x^j \bar{g}(t,x) - D_x^j \bar{g}(s,x) - D_x^j \bar{g}'(s,x)\, \delta \bar{Y}_{s,t}.$$

“Distance” between stochastic controlled vector fields

Given $Y, \bar{Y} \in C^{\alpha}([0,T];\mathbb{R}^{d_Y})$ and $(g,g') \in \mathcal{D}_Y^{2\alpha} C_b^{\beta}$, $(\bar{g},\bar{g}') \in \mathcal{D}_{\bar{Y}}^{2\alpha} C_b^{\beta}$ $\beta = N+\kappa > 1$, we denote by

$$[\![g, g'; \bar{g}, \bar{g}']\!]_{Y,\bar{Y};2\alpha;\beta} := \sum_{i=0}^{N} [\![\delta D_x^i (g-\bar{g})]\!]_{\alpha} + \sum_{j=0}^{N-1} \left([\![\delta D_x^j (g'-\bar{g}')]\!]_{\alpha} + [\![R^{D_x^j g} - \bar{R}^{D_x^j \bar{g}}]\!]_{2\alpha} \right),$$

with

$$\bar{R}_{s,t}^{D_x^j \bar{g}}(x) := D_x^j \bar{g}(t,x) - D_x^j \bar{g}(s,x) - D_x^j \bar{g}'(s,x)\,\delta \bar{Y}_{s,t}.$$

Similarly, given $(G,G') \in \mathcal{D}_Y^{2\alpha} L_p \mathrm{Lin}$ and $(\bar{G},\bar{G}') \in \mathcal{D}_{\bar{Y}}^{2\alpha} L_p \mathrm{Lin}$, we denote by

$$[\![G, G'; \bar{G}, \bar{G}']\!]_{Y,\bar{Y};2\alpha;p}^{\mathrm{Lin}} := \sup_{0 \le s < t \le T} \left(\frac{\|\delta(G-\bar{G})_{s,t}\|_{\mathrm{Lin},p}}{|t-s|^{\alpha}} + \frac{\|\delta(G'-\bar{G}')_{s,t}\|_{\mathrm{Lin},p}}{|t-s|^{\alpha}} + \frac{\|R_{s,t}^{G} - \bar{R}_{s,t}^{\bar{G}}\|_{\mathrm{Lin},p}}{|t-s|^{2\alpha}} \right),$$

with

$$\|\cdot\|_{\mathrm{Lin},p} := \left(\mathbb{E}(|\cdot|_{\mathrm{Lin}}^{p}) \right)^{1/p}, \qquad R_{s,t}^{G} x := G_t x - G_s x - G_s' x\,\delta Y_{s,t}, \quad \bar{R}_{s,t}^{\bar{G}} x := \bar{G}_t x - \bar{G}_s x - \bar{G}_s' x\,\delta \bar{Y}_{s,t}.$$

Assumption R – Distances of the coefficients are controlled by the rough path distance

There is a positive constant $C > 0$ such that, for any $\mathbf{Y}, \bar{\mathbf{Y}} \in \mathcal{C}^{0,\alpha,1}$, the following hold:

- $\sup_{\mathbf{Z}\in\mathcal{C}^{0,\alpha,1}} \|b(\cdot,\cdot;\mathbf{Z})\|_{\mathrm{Lip}} < +\infty$ and $\sup_{t\in[0,T]} \sup_{x\in\mathbb{R}^{d_X}} |b(t,x;\mathbf{Y}) - b(t,x;\bar{\mathbf{Y}})| \le C\rho_\alpha(\mathbf{Y},\bar{\mathbf{Y}})$,

  with similar conditions for σ;

- $\sup_{\mathbf{Z}\in\mathcal{C}^{0,\alpha,1}} \|(f(\cdot,\cdot;\mathbf{Z}), f'(\cdot,\cdot;\mathbf{Z}))\|_\beta + [\![f(\cdot,\cdot;\mathbf{Z}), f'(\cdot,\cdot;\mathbf{Z})]\!]_{Z;2\alpha;\beta} < +\infty$

  and, denoting by $(f,f') = (f(\cdot,\cdot;\mathbf{Y}), f'(\cdot,\cdot;\mathbf{Y}))$ and $(\bar{f},\bar{f}') = (f(\cdot,\cdot;\bar{\mathbf{Y}}), f'(\cdot,\cdot;\bar{\mathbf{Y}}))$,

  $\|(f-\bar{f}, f'-\bar{f}')\|_{\beta-1} + [\![f,f';\bar{f},\bar{f}']\!]_{Y,\bar{Y};2\alpha;\beta-1} \le C\rho_\alpha(\mathbf{Y},\bar{\mathbf{Y}})$;

- $[t \mapsto (h(t,\cdot;\mathbf{Y}), h'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_Y^{2\alpha} C_b^3$ with

  $\sup_{\mathbf{Z}\in\mathcal{C}^{0,\alpha,1}} \|(h(\cdot,\cdot;\mathbf{Z}), h'(\cdot,\cdot;\mathbf{Z}))\|_3 + [\![h(\cdot,\cdot;\mathbf{Z}), h'(\cdot,\cdot;\mathbf{Z})]\!]_{Z;2\alpha;3} < +\infty$,

  and, denoting by $(h,h') = (h(\cdot,\cdot;\mathbf{Y}), h'(\cdot,\cdot;\mathbf{Y}))$ and $(\bar{h},\bar{h}') = (h(\cdot,\cdot;\bar{\mathbf{Y}}), h'(\cdot,\cdot;\bar{\mathbf{Y}}))$,

  $\|(h-\bar{h}, h'-\bar{h}')\|_1 + [\![h,h';\bar{h},\bar{h}']\!]_{Y,\bar{Y};2\alpha;2} \le C\rho_\alpha(\mathbf{Y},\bar{\mathbf{Y}})$;

- $\|X_0^{\mathbf{Y}} - X_0^{\bar{\mathbf{Y}}}\|_2 \le C\rho_\alpha(\mathbf{Y},\bar{\mathbf{Y}})$.

Lemma 7

Under Assumption E and Assumption R, for any $R > 0$ and for any $\mathbf{Y}, \bar{\mathbf{Y}} \in (\mathcal{C}^{0,\alpha,1}, \rho_\alpha)$ with $|\!|\!|\mathbf{Y}|\!|\!|_\alpha \vee |\!|\!|\bar{\mathbf{Y}}|\!|\!|_\alpha \leq R$, there exists a constant $K = K_R > 0$ such that

$$\sup_{t\in[0,T]} \|X_t^{\mathbf{Y}} - X_t^{\bar{\mathbf{Y}}}\|_2 \leq K \rho_\alpha(\mathbf{Y}, \bar{\mathbf{Y}}) \quad \textit{and} \quad \sup_{t\in[0,T]} \|Z_t^{\mathbf{Y}} - Z_t^{\bar{\mathbf{Y}}}\|_1 \leq K \rho_\alpha(\mathbf{Y}, \bar{\mathbf{Y}}).$$

Theorem 8

Under Assumption E and Assumption R, the following statements hold, uniformly in $t \in [0, T]$:

(1) *for any fixed $\varphi : \mathbb{R}^{d_X} \to \mathbb{R}$ bounded and Lipschitz, the map $\mathbf{Y} \mapsto \mu_t^{\mathbf{Y}}(\varphi)$ is locally Lipschitz in $\mathbf{Y} \in (\mathcal{C}^{0,\alpha,1}, \rho_\alpha)$;*

(2) *for any fixed $\varphi : \mathbb{R}^{d_X} \to \mathbb{R}$ bounded and Lipschitz, the map $\mathbf{Y} \mapsto \varsigma_t^{\mathbf{Y}}(\varphi)$ is locally Lipschitz in $\mathbf{Y} \in (\mathcal{C}^{0,\alpha,1}, \rho_\alpha)$;*

(3) *the map $\mathbf{Y} \mapsto \mu_t^{\mathbf{Y}} \in \mathcal{M}_F(\mathbb{R}^{d_X})$ is continuous, if $\mathcal{M}_F(\mathbb{R}^{d_X})$ is equipped with the topology of weak convergence;*

(4) *the map $\mathbf{Y} \mapsto \varsigma_t^{\mathbf{Y}} \in \mathcal{P}(\mathbb{R}^{d_X})$ is continuous, if $\mathcal{P}(\mathbb{R}^{d_X})$ is equipped with the topology of weak convergence.*

Proof of Theorem 8 – Point (1)

Denote by $|\varphi|_{Lip}$ its Lipschitz constant and let $|\varphi|_\infty := \sup_{x\in\mathbb{R}^{d_X}} |\varphi(x)| < +\infty$:

$$
\begin{aligned}
\text{(def)} \quad & |\mu_t^{\mathbf{Y}}(\varphi) - \mu_t^{\bar{\mathbf{Y}}}(\varphi)| = |\mathbb{E}\big[\varphi(X_t^{\mathbf{Y}})Z_t^{\mathbf{Y}} - \varphi(X_t^{\bar{\mathbf{Y}}})Z_t^{\bar{\mathbf{Y}}}\big]| \\
(\pm\, \varphi(X_t^{\mathbf{Y}})Z_t^{\bar{\mathbf{Y}}} \text{ and } \triangle\text{-ineq.}) \quad & \leq \mathbb{E}\big(|\varphi(X_t^{\mathbf{Y}}) - \varphi(X_t^{\bar{\mathbf{Y}}})|Z_t^{\mathbf{Y}}\big) + \mathbb{E}\big(|\varphi(X_t^{\bar{\mathbf{Y}}})||Z_t^{\mathbf{Y}} - Z_t^{\bar{\mathbf{Y}}}|\big) \\
\text{(Hölder's ineq.)} \quad & \leq \|\varphi(X_t^{\mathbf{Y}}) - \varphi(X_t^{\bar{\mathbf{Y}}})\|_2 \|Z_t^{\mathbf{Y}}\|_2 + \mathbb{E}\big(|\varphi(X_t^{\bar{\mathbf{Y}}})||Z_t^{\mathbf{Y}} - Z_t^{\bar{\mathbf{Y}}}|\big) \\
\text{(Lipschitz \& bounded } \varphi) \quad & \leq |\varphi|_{Lip}\|X_t^{\mathbf{Y}} - X_t^{\bar{\mathbf{Y}}}\|_2 \|Z_t^{\mathbf{Y}}\|_2 + |\varphi|_\infty \|Z_t^{\mathbf{Y}} - Z_t^{\bar{\mathbf{Y}}}\|_1 \\
\text{(Lemma 7)} \quad & \lesssim |\varphi|_{Lip}\rho_\alpha(\mathbf{Y},\bar{\mathbf{Y}})\|Z_t^{\mathbf{Y}}\|_2 + |\varphi|_\infty \rho_\alpha(\mathbf{Y},\bar{\mathbf{Y}}).
\end{aligned}
$$

We conclude recalling that $\mathbf{Y} \mapsto e^{I_t^{\mathbf{Y}}} = Z_t^{\mathbf{Y}}$ is locally bounded.

Note that $\mu_t^{\mathbf{Y}}(1) = \mathbb{E}(e^{I_t^{\mathbf{Y}}}) > 0$. By $(i.1)$, $\mathbf{Y} \mapsto \mu_t^{\mathbf{Y}}(1)$ is continuous and therefore it is locally uniformly bounded away from zero. Noting that

$$\text{(def)} \quad |\varsigma_t^{\mathbf{Y}}(\varphi) - \varsigma_t^{\bar{\mathbf{Y}}}(\varphi)| = \left| \frac{\mu_t^{\mathbf{Y}}(\varphi)}{\mu_t^{\mathbf{Y}}(1)} - \frac{\mu_t^{\bar{\mathbf{Y}}}(\varphi)}{\mu_t^{\bar{\mathbf{Y}}}(1)} \right|$$

$$\left(\pm \frac{\mu_t^{\bar{\mathbf{Y}}}(\varphi)}{\mu_t^{\mathbf{Y}}(1)} \text{ and } \triangle\text{-ineq.}\right) \quad \leq \frac{1}{|\mu_t^{\mathbf{Y}}(1)|} \left|\mu_t^{\mathbf{Y}}(\varphi) - \mu_t^{\bar{\mathbf{Y}}}(\varphi)\right| + \frac{|\mu_t^{\bar{\mathbf{Y}}}(\varphi)|}{|\mu_t^{\mathbf{Y}}(1)\mu_t^{\bar{\mathbf{Y}}}(1)|} \left|\mu_t^{\mathbf{Y}}(1) - \mu_t^{\bar{\mathbf{Y}}}(1)\right|,$$

we conclude using the previous point.

Let $(\mathbf{Y}^n)_n$ be a sequence in $\mathcal{C}^{0,\alpha,1}$ such that $\rho_\alpha(\mathbf{Y}^n, \mathbf{Y}) \to 0$ as $n \to +\infty$. Let $\varphi \in C_b^0(\mathbb{R}^{d_x}; \mathbb{R})$ be arbitrary.

By the continuous mapping theorem, since φ is continuous and $X_t^{\mathbf{Y}^n} \to X_t^{\mathbf{Y}}$ in L_2 (Lemma 7), we have $\varphi(X_t^{\mathbf{Y}^n}) \to \varphi(X_t^{\mathbf{Y}})$ in probability. Moreover, $|\varphi(X_t^{\mathbf{Y}^n}) - \varphi(X_t^{\mathbf{Y}})|^2 \leq 4\|\varphi\|_\infty^2$, which provides an integrable dominating function. Therefore, by the dominated convergence theorem, it follows that

$$\mathbb{E}\Big[\,|\varphi(X_t^{\mathbf{Y}^n}) - \varphi(X_t^{\mathbf{Y}})|^2\Big] \to 0, \quad \text{i.e.} \quad \varphi(X_t^{\mathbf{Y}^n}) \to \varphi(X_t^{\mathbf{Y}}) \text{ in } L_2.$$

Hence:

$$\begin{aligned}
\text{(def)} \quad & |\mu_t^{\mathbf{Y}^n}(\varphi) - \mu_t^{\mathbf{Y}}(\varphi)| = \big|\mathbb{E}\big[\varphi(X_t^{\mathbf{Y}^n}) Z_t^{\mathbf{Y}^n} - \varphi(X_t^{\mathbf{Y}}) Z_t^{\mathbf{Y}}\big]\big| \\
(\pm\, \varphi(X_t^{\mathbf{Y}^n}) Z_t^{\mathbf{Y}} \text{ and } \triangle\text{-ineq.}) \quad & \leq \mathbb{E}\big(|\varphi(X_t^{\mathbf{Y}^n}) - \varphi(X_t^{\mathbf{Y}})| Z_t^{\mathbf{Y}}\big) + \mathbb{E}\big(|\varphi(X_t^{\mathbf{Y}^n})|\,|Z_t^{\mathbf{Y}^n} - Z_t^{\mathbf{Y}}|\big) \\
\text{(Hölder's ineq.)} \quad & \leq \|\varphi(X_t^{\mathbf{Y}^n}) - \varphi(X_t^{\mathbf{Y}})\|_2\, \|Z_t^{\mathbf{Y}}\|_2 + \mathbb{E}\big(|\varphi(X_t^{\mathbf{Y}^n})|\,|Z_t^{\mathbf{Y}^n} - Z_t^{\mathbf{Y}}|\big) \\
(\text{bounded } \varphi) \quad & \leq \|\varphi(X_t^{\mathbf{Y}^n}) - \varphi(X_t^{\mathbf{Y}})\|_2\, \|Z_t^{\mathbf{Y}}\|_2 + |\varphi|_\infty\, \|Z_t^{\mathbf{Y}^n} - Z_t^{\mathbf{Y}}\|_1 \\
\text{(Lemma 7)} \quad & \longrightarrow 0, \quad \text{as } n \to +\infty.
\end{aligned}$$

$\square$

Rough stochastic filtering

Rough Zakai equation in spaces of measures: existence

Let us set the random operator:

$$A_t\varphi(x) = \sum_{i=1}^{d_X} \partial_i\varphi(x)\,\tilde{b}^i(t,x,Y_t) + \frac{1}{2}\sum_{i,j=1}^{d_X} \partial^2_{ij}\varphi(x)\,\tilde{a}^{ij}(t,x,Y_t),$$

$$\tilde{a}^{ij}(t,x,Y_t) = \sum_{\theta=1}^{d_B} \sigma^i_\theta\sigma^j_\theta(t,x,Y_t) + \sum_{\lambda,\eta=1}^{d_Y} f^i_\lambda(t,x,Y_t)\,\frac{d\langle Y^\lambda, Y^\eta\rangle_t}{dt}\,f^j_\eta(t,x,Y_t),$$

$$\tilde{b}^i(t,x,Y_t) = b^i(t,x,Y_t) + \sum_{\lambda,\eta=1}^{d_Y} f^i_\lambda(t,x,Y_t)\,\frac{d\langle Y^\lambda, Y^\eta\rangle_t}{dt}\,h_\eta(t,x,Y_t).$$

Theorem 9 ([CP24], Theorem 3.11 and Theorem 5.5)

Under Assumption CP-E (in the paper) , the unnormalised stochastic filter μ_t satisfy the stochastic differential equation

$$\mu_t(\varphi) = \mu_0(\varphi) + \int_0^t \mu_r(A_r\varphi)\,dr + \int_0^t \mu_r\big(\nabla\varphi\, f(r,\cdot,Y_r) + \varphi\, h(r,\cdot,Y_r)\big)\,dY_r, \qquad \mu_0 = \mathrm{Law}(X_0).$$

for any $\varphi \in C_b^2(\mathbb{R}^{d_X};\mathbb{R})$.
Morover, under Assumption CP-E and Assumption CP-U (in the paper), it is the unique $(\mathcal{F}_t^Y)_t$–adapted solution of the above Zakai equation satisfyng $\mathbb{E}[\sup_{t\in[0,T]} \mu(1)^2] < +\infty$ for any $T > 0$.

Analogously to the stochastic case, we aim to give meaning to an equation of the form

$$\mu_t^{\mathbf{Y}}(\varphi) = \mu_0^{\mathbf{Y}}(\varphi) + \int_0^t \mu_r(A_r^{\mathbf{Y}}\varphi)\,dr + \int_0^t \mu_r^{\mathbf{Y}}(\Gamma_r^{\mathbf{Y}}\varphi)\,d\mathbf{Y}_r, \qquad \mu_0^{\mathbf{Y}} = \mathrm{Law}(X_0^{\mathbf{Y}}), \tag{4}$$

where

$$A_t^{\mathbf{Y}}\varphi(x) = \sum_{i=1}^{d_X} \left(\bar{b}^{[\mathbf{Y}]}\right)^i(t,x;\mathbf{Y})\,\partial_i\varphi(x) + \frac{1}{2}\sum_{i,j=1}^{d_X} \left(\bar{a}^{[\mathbf{Y}]}\right)^{ij}(t,x;\mathbf{Y})\,\partial_{ij}^2\varphi(x),$$

$$\left(\bar{a}^{[\mathbf{Y}]}\right)^{ij}(t,x;\mathbf{Y}) := \sum_{\theta=1}^{d_B} \sigma_\theta^i\sigma_\theta^j(t,x;\mathbf{Y}) + \sum_{\kappa,\lambda=1}^{d_Y} f_\kappa^i(t,x;\mathbf{Y})\,[\dot{\mathbf{Y}}]_t^{\kappa\lambda}\,f_\lambda^j(t,x;\mathbf{Y}),$$

$$\left(\bar{b}^{[\mathbf{Y}]}\right)^i(t,x;\mathbf{Y}) := b^i(t,x;\mathbf{Y}) + \sum_{\kappa,\lambda=1}^{d_Y} f_\kappa^i(t,x;\mathbf{Y})\,[\dot{\mathbf{Y}}]_t^{\kappa\lambda}\,h_\lambda(t,x;\mathbf{Y}),$$

$$(\Gamma_t^{\mathbf{Y}})_\kappa\varphi(x) = \sum_{i=1}^{d_X} f_\kappa^i(t,x;\mathbf{Y})\,\partial_i\varphi(x) + h_\kappa(t,x;\mathbf{Y})\,\varphi(x),$$

$$(\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda}\varphi(x) := \sum_{i=1}^{d_X} (f'_{\kappa\lambda})^i(t,x;\mathbf{Y})\,\partial_i\varphi(x) + h'_{\kappa\lambda}(t,x;\mathbf{Y})\,\varphi(x).$$

Rough Zakai equation (definition)

A measure-valued solution to Equation (4) is a curve $\mu : [0,T] \to \mathcal{M}_F(\mathbb{R}^{d_X})$ weakly continuos (for any $\varphi \in C_b^0(\mathbb{R}^{d_x};\mathbb{R})$, the map $t \mapsto \langle \mu_t, \varphi \rangle$ is continuous) such that for any $\varphi \in C_b^3(\mathbb{R}^{d_x};\mathbb{R})$, they hold:

(i) $t \mapsto \mu_t(A_t^{\mathbf{Y}}\varphi) \in L_1([0,T], dt)$ (the middle integral is a well-defined Lebesgue integral).

(ii) There exists $C = C_\varphi > 0$, uniform on bounded sets of φ, such that for any $0 \le s \le t \le T$ and any $\kappa, \lambda = 1, \dots, d_Y$,

$$\left|\mu_t\Big((\Gamma_t^{\mathbf{Y}})_\kappa(\Gamma_t^{\mathbf{Y}})_\lambda\varphi + (\Gamma_t^{\mathbf{Y}})'_{\lambda\kappa}\varphi\Big) - \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda\varphi + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa}\varphi\Big)\right| \le C_\varphi|t-s|^\alpha,$$

$$\left|\mu_t((\Gamma_t^{\mathbf{Y}})_\kappa\varphi) - \mu_s((\Gamma_s^{\mathbf{Y}})_\kappa\varphi) - \sum_{\eta=1}^{d_Y}\mu_s\Big((\Gamma_s^{\mathbf{Y}})_\eta(\Gamma_s^{\mathbf{Y}})_\kappa\varphi + (\Gamma_s^{\mathbf{Y}})'_{\kappa\eta}\varphi\Big)\,\delta Y_{s,t}^\eta\right| \le C_\varphi|t-s|^{2\alpha}.$$

(iii) (Davie-type expansion) For any $0 \le s \le t \le T$,

$$\mu_t(\varphi) - \mu_s(\varphi) = \int_s^t \mu_r(A_r^{\mathbf{Y}}\varphi)\,dr + \sum_{\kappa=1}^{d_Y}\mu_s((\Gamma_s^{\mathbf{Y}})_\kappa\varphi)\,\delta Y_{s,t}^\kappa$$
$$+ \sum_{\kappa,\lambda=1}^{d_Y}\mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda\varphi + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa}\varphi\Big)\,\mathbb{Y}_{s,t}^{\kappa\lambda} + \mu_{s,t}^{\varphi,\flat},$$

where $\mu_{s,t}^{\varphi,\flat} = o(|t-s|)$ as $|t-s| \to 0$.

Lemma 10

Let $\mu : [0, T] \to \mathcal{M}_F(\mathbb{R}^{d_x})$ *satisfy condition* (ii) *of the previous definition. Then, for any* $t \in [0, T]$*, the following limit exists along any sequence of partitions* π *of* $[0, t]$ *whose mesh tends to zero:*

$$\int_0^t \mu_r(\Gamma_r^{\mathbf{Y}}\varphi)\, d\mathbf{Y}_r \;:=\; \lim_{|\pi|\to 0} \sum_{[s,u]\in\pi} \Big(\mu_s\big(((\Gamma_s^{\mathbf{Y}})_\kappa\varphi\big)\, \delta Y_{s,u}^{\kappa} \;+\; \mu_s\big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda\varphi + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa}\varphi\big)\, \mathbb{Y}_{s,u}^{\kappa\lambda}\Big),$$

with implicit summation over $\kappa, \lambda = 1, \ldots, d_Y$.

In particular, condition (iii) *of the previous definition can be equivalently replaced by*

(iii)′ *for any* $t \in [0, T]$,

$$\mu_t(\varphi) = \mu_0(\varphi) + \int_0^t \mu_r(A_r^{\mathbf{Y}}\varphi)\, dr + \int_0^t \mu_r(\Gamma_r^{\mathbf{Y}}\varphi)\, d\mathbf{Y}_r,$$

and the quantity $\mu_{s,t}^{\varphi,\flat}$ *in fact satisfies*

$$\big|\mu_{s,t}^{\varphi,\flat}\big| \leq K_\varphi\, |t-s|^{3\alpha} \qquad \textit{for any } s, t \in [0, T],$$

where $K_\varphi > 0$ *is uniform over bounded sets of* φ *in* $C_b^3(\mathbb{R}^{d_x}; \mathbb{R})$.

Let $t \in [0, T]$. For any $\varphi \in C_b^3(\mathbb{R}^{d_x}; \mathbb{R})$ and for any $s, u \in [0, t]$, define

$$\Xi^{\varphi}_{s,u} := \langle \mu_s, (\Gamma^{\mathbf{Y}}_s)_\kappa \varphi \rangle \, \delta Y^{\kappa}_{s,u} + \langle \mu_s, ((\Gamma^{\mathbf{Y}}_s)_\kappa (\Gamma^{\mathbf{Y}}_s)_\lambda + (\Gamma^{\mathbf{Y}}_s)'_{\lambda\kappa}) \varphi \rangle \, \mathbb{Y}^{\kappa\lambda}_{s,u}.$$

By Chen's relation and condition (ii), it follows straightforwardly that

$$|\Xi^{\varphi}_{s,v} - \Xi^{\varphi}_{s,u} - \Xi^{\varphi}_{u,v}| \leq C_\varphi \, \|\mathbf{Y}\|_\alpha \, |t - s|^{3\alpha},$$

with C_φ uniform over bounded sets of $\varphi \in C_b^3(\mathbb{R}^{d_x}; \mathbb{R})$.

Hence, thanks to the classical sewing lemma the proof is completed.

$\square$

The unnormalised rough filter is a solution of the rough Zakai equation

Theorem 11

Under Assumption E, the path

$$[0,T] \ni t \longmapsto \mu_t^{\mathbf{Y}} \in \mathcal{M}_{\mathcal{F}}(\mathbb{R}^{d_X})$$

given by the unnormalised rough filter is a measure–valued solution to Equation (4)*, starting from*

$$\mu_0^{\mathbf{Y}} = \mathrm{Law}(X_0^{\mathbf{Y}}).$$

Moreover, for any $R > 0$ and for any $\theta \in (0,1]$,

$$\sup_{\substack{\phi \in C_b^\theta(\mathbb{R}^{d_X};\mathbb{R}) \\ \|\phi\|_{C_b^\theta} \leq R}} \sup_{0 \leq s < t \leq T} \frac{|\mu_t^{\mathbf{Y}}(\phi) - \mu_s^{\mathbf{Y}}(\phi)|}{|t-s|^{\theta\alpha}} < +\infty.$$

Proof of Theorem 11

Firstly, we consider the case $h \equiv 0 \Rightarrow Z_t^{\mathbf{Y}} \equiv 1$. The unnormalised filter is

$$\mu_t^{\mathbf{Y}}(\varphi) = \mathbb{E}\left[\varphi\left(X_t^{\mathbf{Y}}\right)\right], \qquad \varphi \in C_b^3(\mathbb{R}^{d_X}),$$

- μ^Y **is weakly continuous.** Let $s \to t$. By definition of $L_{2,\infty}$, $X^{\mathbf{Y}}$ is a.s-continuous so:

  $$X_s^{\mathbf{Y}} \xrightarrow{a.s.} X_t^{\mathbf{Y}} \quad \Rightarrow \quad X_s^{\mathbf{Y}} \xrightarrow{\mathcal{L}} X_t^{\mathbf{Y}}.$$

  Hence:

  $$\mu_t^{\mathbf{Y}}(\varphi) = \mathbb{E}[\varphi(X_t^{\mathbf{Y}})] \longrightarrow \mathbb{E}[\varphi(X_s^{\mathbf{Y}})] = \mu_s^{\mathbf{Y}}(\varphi).$$

- **Measurability and uniform boundedness of $t \mapsto \mu_t^Y(A_t^Y\varphi)$.**
  - By Assumption E, $(t, x) \mapsto A_t^{\mathbf{Y}}\varphi(x)$ is continuous; composing with the continuous path $t \mapsto X_t^{\mathbf{Y}}$ gives that $t \mapsto A_t^{\mathbf{Y}}\varphi(X_t^{\mathbf{Y}})$ is measurable, hence for Fubini's theorem $t \mapsto \mu_t^{\mathbf{Y}}(A_t^{\mathbf{Y}}\varphi)$ is measurable.
  - Using linear growth of b, σ, f and bounded $[\dot{\mathbf{Y}}]$,

    $$|A_t^{\mathbf{Y}}\varphi(x)| \;\leq\; C_{\mathbf{Y}}(1 + |x|^2)\|\varphi\|_{C_b^2}$$

    Since $X^Y \in L_{2,\infty}$, $\sup_{t\in[0,T]} \mathbb{E}[|X_t^{\mathbf{Y}}|^2] < \infty$, hence

    $$\sup_{t\in[0,T]} |\mu_t^{\mathbf{Y}}(A_t^{\mathbf{Y}}\varphi)| \;\leq\; C_{\mathbf{Y}}\,\|\varphi\|_{C_b^2}\,\sup_t \mathbb{E}[1 + |X_t^{\mathbf{Y}}|^2] \;<\; \infty.$$

- **Condition** (ii).
  Since $(X^{\mathbf{Y}}_{\cdot}, f(\cdot, X^{\mathbf{Y}}, \mathbf{Y})) \in \mathbf{D}^{2\alpha}_{Y} L_p$ for $p \in [2, \infty)$, we can expand as it follows controlling the remainders:

$$(\Gamma^{\mathbf{Y}}_t)_\kappa(\Gamma^{\mathbf{Y}}_t)_\lambda\varphi(X^{\mathbf{Y}}_t) + (\Gamma^{\mathbf{Y}}_t)'_{\lambda\kappa}\varphi(X^{\mathbf{Y}}_t) - \left[(\Gamma^{\mathbf{Y}}_s)_\kappa(\Gamma^{\mathbf{Y}}_s)_\lambda\varphi(X^{\mathbf{Y}}_s) + (\Gamma^{\mathbf{Y}}_s)'_{\lambda\kappa}\varphi(X^{\mathbf{Y}}_s)\right] = Q^{\varphi}_{s,t},$$

$$(\Gamma^{\mathbf{Y}}_t)_\kappa\varphi(X^{\mathbf{Y}}_t) - (\Gamma^{\mathbf{Y}}_s)_\kappa\varphi(X^{\mathbf{Y}}_s) - \sum_\eta \left[(\Gamma^{\mathbf{Y}}_s)_\eta(\Gamma^{\mathbf{Y}}_s)_\kappa\varphi(X^{\mathbf{Y}}_s) + (\Gamma^{\mathbf{Y}}_s)'_{\kappa\eta}\varphi(X^{\mathbf{Y}}_s)\right]\delta Y^{\eta}_{s,t} = R^{\varphi}_{s,t},$$

with $||Q^{\varphi}_{s,t}||_2 \leq c_\varphi|t-s|^\alpha$ and $||\mathbb{E}[R^{\varphi}_{s,t}|\mathcal{F}_s]||_2 \leq c_\varphi|t-s|^{2\alpha}$.

Hence, in mean value, since $|\mathbb{E}[Q^{\varphi}_{s,t}]| \leq ||Q^{\varphi}_{s,t}||_2$ and $|\mathbb{E}[R^{\varphi}_{s,t}]| \leq ||\mathbb{E}[R^{\varphi}_{s,t}|\mathcal{F}_s]||_2$, we have

$$\boxed{\left|\,\mu^{\mathbf{Y}}_t\left((\Gamma^{\mathbf{Y}}_t)_\kappa(\Gamma^{\mathbf{Y}}_t)_\lambda\varphi + (\Gamma^{\mathbf{Y}}_t)'_{\lambda\kappa}\varphi\right) - \mu^{\mathbf{Y}}_s\left((\Gamma^{\mathbf{Y}}_s)_\kappa(\Gamma^{\mathbf{Y}}_s)_\lambda\varphi + (\Gamma^{\mathbf{Y}}_s)'_{\lambda\kappa}\varphi\right)\right| \lesssim |t-s|^\alpha.}$$

$$\boxed{\left|\mu^{\mathbf{Y}}_t((\Gamma^{\mathbf{Y}}_t)_\kappa\varphi) - \mu^{\mathbf{Y}}_s((\Gamma^{\mathbf{Y}}_s)_\kappa\varphi) - \sum_\eta \mu^{\mathbf{Y}}_s\Big((\Gamma^{\mathbf{Y}}_s)_\eta(\Gamma^{\mathbf{Y}}_s)_\kappa\varphi + (\Gamma^{\mathbf{Y}}_s)'_{\kappa\eta}\varphi\Big)\delta Y^{\eta}_{s,t}\right| \lesssim |t-s|^{2\alpha}.}$$

- **Condition** (iii)**.** Consider

$$dX_t = b_t\,dt + \sigma_t\,dB_t + f_t\,d\mathbf{Y}_t.$$

Then, for any φ under suitable regularity assumptions, it holds the **rough Itô formula** ([FHL24], Theorem 4.3):

$$\boxed{\begin{aligned}\delta\varphi(X)_{s,t} = {} & \int_s^t \partial_i\varphi(X_r)\,b^i(r,X_r)\,dr + \int_s^t \partial_i\varphi(X_r)\,\sigma^i_\theta(r,X_r)\,dB^\theta_r \\ & + \partial_i\varphi(X_s)\,f^i_\kappa(s,X_s)\,\delta Y^\kappa_{s,t} \\ & + \left[\partial^2_{ij}\varphi(X_s)\,f^i_\kappa(s,X_s)\,f^j_\lambda(s,X_s) + \partial_i\varphi(X_s)\,(f'_{\kappa\lambda})_i(s,X_s)\right]\mathbb{Y}^{\kappa\lambda}_{s,t} \\ & + \tfrac{1}{2}\int_s^t \partial^2_{ij}\varphi(X_r)\Big(\sigma^i_\theta\sigma^j_\theta + f^i_\kappa\,[\dot{Y}]^{\kappa\lambda}_r\,f^j_\lambda\Big)(r,X_r)\,dr + \varphi^\natural_{s,t}\end{aligned}}$$

and the remainder satisfies

$$||\mathbb{E}[\varphi^\natural_{s,t}|\mathcal{F}_s]||_2 = O(|t-s|^{3\alpha}).$$

So, we just apply the formula to $\varphi(X^{\mathbf{Y}})$, take the expected value and see that the terms coming from the Brownian motion disappear, since they are expected value of martingales starting from zero.

The proof when h is not zero follows as a generalization of the previous argument. Indeed, by the product formula ([BFS25], Proposition 4.9), we conclude that, for any $0 \le s \le t \le T$,

$$\varphi(X_t^{\mathbf{Y}})Z_t^{\mathbf{Y}} - \varphi(X_s^{\mathbf{Y}})Z_s^{\mathbf{Y}} = \int_s^t A_r^{\mathbf{Y}}\varphi(X_r^{\mathbf{Y}})\, Z_r^{\mathbf{Y}}\, dr + \sum_{\theta=1}^{d_B}\int_s^t \partial_j\varphi(X_r^{\mathbf{Y}})\,\sigma_{j\theta}(r, X_r^{\mathbf{Y}}; \mathbf{Y})\, Z_r^{\mathbf{Y}}\, dB_r^{\theta}$$

$$+ \sum_{\kappa=1}^{d_Y}(\Gamma_s^{\mathbf{Y}})_\kappa\varphi(X_s^{\mathbf{Y}})\, Z_s^{\mathbf{Y}}\,\delta Y_{s,t}^{\kappa}$$

$$+ \sum_{\kappa,\lambda=1}^{d_Y}\Big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda\varphi(X_s^{\mathbf{Y}}) + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa}\varphi(X_s^{\mathbf{Y}})\Big) Z_s^{\mathbf{Y}}\,\mathbb{Y}_{s,t}^{\kappa\lambda} + \varphi_{s,t}^{\natural},$$

where

$$\mathbb{E}\big[|\varphi_{s,t}^{\natural}| \mid \mathcal{F}_s\big] \;\le\; c_\varphi\,|t-s|^{3\alpha} \quad \text{for some constant } c_\varphi > 0,$$

uniformly on bounded sets in $C_b^3(\mathbb{R}^{d_X}; \mathbb{R})$.

Taking expectations on both sides and recalling that $\mu_t^Y(\varphi) = \mathbb{E}[\varphi(X_t^Y)Z_t^Y]$ yields the rough Zakai equation. □

Rough stochastic filtering

Rough Zakai equation in spaces of measures: uniqueness

Backward Rough PDE (definition)

A function-valued solution to the backward equation

$$u_t(x) = u_T(x) + \int_t^T \left(A_r^{\mathbf{Y}} u_r(x) - (\Gamma_r^{\mathbf{Y}}) u_r(x)\,[\dot{\mathbf{Y}}_r]\right) dr + \int_t^T \Gamma_r^{\mathbf{Y}} u_r(x)\, d\mathbf{Y}_r \tag{5}$$

on the time interval $[0, T]$ is a map $u : [0, T] \times \mathbb{R}^d \to \mathbb{R}$ satisfying the following properties:

(i) for any $x \in \mathbb{R}^d$, the function

$$t \mapsto A_t^{\mathbf{Y}} u_t(x) - \sum_{\kappa=1}^{d_{\mathbf{Y}}} (\Gamma_t^{\mathbf{Y}})_\kappa u_t(x)\,[\dot{\mathbf{Y}}_t]^\kappa$$

belongs to $L_1([0, T], dt)$;

(ii) there is a constant $C > 0$ such that, for any $0 \le s \le t \le T$ and for any $\kappa, \lambda = 1, \dots, d_{\mathbf{Y}}$,

$$\sup_{x\in\mathbb{R}^d} \left| (\Gamma_s^{\mathbf{Y}})_\kappa (\Gamma_s^{\mathbf{Y}})_\lambda u_s(x) - (\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_s(x) - \Big((\Gamma_t^{\mathbf{Y}})_\kappa (\Gamma_t^{\mathbf{Y}})_\lambda u_t(x) - (\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t(x) \Big) \right| \le C|t-s|^\alpha,$$

$$\sup_{x\in\mathbb{R}^d} \left| (\Gamma_s^{\mathbf{Y}})_\kappa u_s(x) - (\Gamma_t^{\mathbf{Y}})_\kappa u_t(x) - \sum_{\eta=1}^{d_{\mathbf{Y}}} \Big((\Gamma_t^{\mathbf{Y}})_\kappa (\Gamma_t^{\mathbf{Y}})_\eta u_t(x) - (\Gamma_t^{\mathbf{Y}})'_{\kappa\eta} u_t(x) \Big) \delta Y_{s,t}^\eta \right| \le C|t-s|^{2\alpha};$$

(iii) for any $0 \leq s \leq t \leq T$ and for any $x \in \mathbb{R}^d$, the following Davie-type expansion holds:

$$u_s(x) - u_t(x) = \int_s^t \left(A_r^{\mathbf{Y}} u_r(x) - \sum_{\kappa=1}^{d_{\mathbf{Y}}} (\Gamma_t^{\mathbf{Y}})'_{\lambda\kappa} u_r(x) [\dot{\mathbf{Y}}_r]^{\kappa\lambda} \right) dr$$

$$+ \sum_{\kappa=1}^{d_{\mathbf{Y}}} (\Gamma_t^{\mathbf{Y}})_\kappa u_t(x) \, \delta Y_{s,t}^\kappa + \sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} \Big((\Gamma_t^{\mathbf{Y}})_\kappa (\Gamma_t^{\mathbf{Y}})_\lambda u_t(x) - (\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t(x) \Big) \mathbb{Y}_{s,t}^{\kappa,\lambda} + u_{s,t}^\natural(x),$$

with

$$\sup_{x \in \mathbb{R}^{d_X}} \frac{|u_{s,t}^\natural(x)|}{|t-s|^{3\alpha}} < +\infty.$$

When the terminal condition $u_T(\cdot) = g : \mathbb{R}^{d_X} \to \mathbb{R}$ is specified, we say that u is a solution with terminal datum g.

Solution of the rough backward PDE

Assumption U. Let $\theta \in [4,5]$ such that $2\alpha + (\theta - 4)\alpha > 1$. Let the following hold:

- the map $[t \mapsto b(t,\cdot;\mathbf{Y})]$ belongs to

$$C^0([0,T]; C_b^2(\mathbb{R}^{d_X};\mathbb{R}^{d_X})) \cap b\mathcal{B}([0,T]; C_b^3(\mathbb{R}^{d_X};\mathbb{R}^{d_X}));$$

- the map $[t \mapsto \sigma(t,\cdot;\mathbf{Y})]$ belongs to

$$C^0([0,T]; C_b^2(\mathbb{R}^{d};\mathrm{Lin}(\mathbb{R}^{d_B},\mathbb{R}^{d}))) \cap \mathfrak{b}\mathfrak{B}([0,T]; C_b^3(\mathbb{R}^{d};\mathrm{Lin}(\mathbb{R}^{d_B},\mathbb{R}^{d})));$$

- $[t \mapsto (f(t,\cdot;\mathbf{Y}), f'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_{\mathbf{Y}}^{2\alpha} C_b^{\theta}$;
- $[t \mapsto (h(t,\cdot;\mathbf{Y}), h'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_{\mathbf{Y}}^{2\alpha} C_b^{\theta}$.

Note that **Assumption U** implies **Assumption E**.

Solution of the rough backward PDE

Assumption U. Let $\theta \in [4,5]$ such that $2\alpha + (\theta-4)\alpha > 1$. Let the following hold:

- the map $[t \mapsto b(t,\cdot;\mathbf{Y})]$ belongs to
$$C^0([0,T];C_b^2(\mathbb{R}^{d_X};\mathbb{R}^{d_X})) \cap b\mathcal{B}([0,T];C_b^3(\mathbb{R}^{d_X};\mathbb{R}^{d_X}));$$
- the map $[t \mapsto \sigma(t,\cdot;\mathbf{Y})]$ belongs to
$$C^0([0,T];C_b^2(\mathbb{R}^{d};\mathrm{Lin}(\mathbb{R}^{d_B},\mathbb{R}^{d}))) \cap \mathfrak{b}\mathfrak{B}([0,T];C_b^3(\mathbb{R}^{d};\mathrm{Lin}(\mathbb{R}^{d_B},\mathbb{R}^{d})));$$
- $[t \mapsto (f(t,\cdot;\mathbf{Y}), f'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_{\mathbf{Y}}^{2\alpha} C_b^{\theta}$;
- $[t \mapsto (h(t,\cdot;\mathbf{Y}), h'(t,\cdot;\mathbf{Y}))] \in \mathcal{D}_{\mathbf{Y}}^{2\alpha} C_b^{\theta}$.

Note that **Assumption U** implies **Assumption E**.

Theorem 12 ([BFS25], Proposition and Theorem 4.8)

For any final datum $g \in C_b^3(\mathbb{R}^{d_X};\mathbb{R})$, there is a unique function-valued solution u to Equation (5) *which also satisfies*

$$u \in C_b^{0,2}([0,T]\times\mathbb{R}^d;\mathbb{R}) \cap C^{\alpha}([0,T];C_b^2(\mathbb{R}^{d_X};\mathbb{R})) \cap b\mathcal{B}([0,T];C_b^3(\mathbb{R}^{d_X};\mathbb{R})).$$

Theorem 13

Under Assumption U and for any $\nu \in \mathcal{M}_F(\mathbb{R}^{d_X})$, there is at most one solution μ to Equation (4) starting from $\mu_0 = \nu$ such that the following holds, for any $R > 0$:

$$\sup_{\substack{\phi \in C_b^1(\mathbb{R}^{d_x};\mathbb{R}) \\ \|\phi\|_{C_b^1} \le R}} \sup_{0 \le s < t \le T} \frac{|\mu_t(\phi) - \mu_s(\phi)|}{|t-s|^\alpha} < +\infty. \tag{6}$$

Let μ_1, μ_2 be two solutions of Equation (4), both satisfying (6).

Let $\tau \in [0, T]$ and $\varphi \in C_c^\infty(\mathbb{R}^{d_X}; \mathbb{R})$ be fixed but arbitrary; we want to show that $\mu^1_\tau(\varphi) = \mu^2_\tau(\varphi)$.

Let $u = u_t(x) : [0, \tau] \times \mathbb{R}^{d_X} \to \mathbb{R}$ to the backward Equation (5) on $[0, \tau]$ with terminal datum φ.

The conclusion follows if we show that the map is constant for $i = 1, 2$ $t \mapsto \mu^i_t(u_t)$ (it makes sense since $u \in b\mathcal{B}([0, T]; C^3_b(\mathbb{R}^{d_X}; \mathbb{R}))$, indeed we would have

$$\mu^1_\tau(\varphi) = \mu^1_\tau(u_\tau) = \mu^1_0(u_0) = \mu^2_0(u_0) = \mu^2_\tau(u_\tau) = \mu^2_\tau(\varphi).$$

Idea: show that $t \mapsto \mu^i_t(u_t)$ is $(1 + \varepsilon)-$Hölder continuous for some $\varepsilon > 0$.

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \leq s \leq t \leq T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \leq s \leq t \leq T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

First term: u solves the backward Equation (5) so we can use the Davie type expansion for $u_s - u_t$:

$$-\mu_s(u_s - u_t) = -\mu_s\left(\int_s^t \left(A_r^{\mathbf{Y}} u_r - \sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} (\Gamma_r^{\mathbf{Y}})'_{\kappa\lambda} u_r \,[\dot{\mathbf{Y}}_r]^{\kappa\lambda}\right) dr\right)$$

$$-\mu_s\left(\sum_{\kappa=1}^{d_Y} (\Gamma_t^{\mathbf{Y}})_\kappa \, u_t \, \delta Y_{s,t}^{\kappa}\right) - \mu_s\left(\sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} \left((\Gamma_t^{\mathbf{Y}})_\kappa (\Gamma_t^{\mathbf{Y}})_\lambda u_t - (\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t\right) \mathbb{Y}_{s,t}^{\kappa\lambda}\right) + \mu_s(u^\natural_{s,t}),$$

with $|\mu_s(u^\natural_{s,t})| \leq \sup_{r\in[0,\tau]} |\mu_r(1)| \sup_{x\in\mathbb{R}^{d_x}} |u^\natural_{s,t}(x)| \lesssim |t-s|^{3\alpha}$ uniformly in s, t.

Proof of Theorem 13

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \le s \le t \le T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Second term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_s) - \mu_s(u_s)$:

$$+(\mu_t - \mu_s)(u_s) = \mu_t(u_s) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr$$

$$+\sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\, \delta Y_{s,t}^\kappa + \sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa (\Gamma_s^{\mathbf{Y}})_\lambda u_s + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa} u_s\Big)\, \mathbb{Y}_{s,t}^{\kappa\lambda} + \mu_s(u_{s,t}^\natural),$$

$$\text{where } |\mu_s(u_{s,t}^\natural)| \lesssim |t-s|^{3\alpha} \text{ since } \sup_{t\in[0,\tau]} |u_t|_{\mathcal{C}_b^3} < +\infty.$$

Proof of Theorem 13

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \le s \le t \le T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Second term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_s) - \mu_s(u_s)$:

$$+(\mu_t - \mu_s)(u_s) = \mu_t(u_s) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr$$

$$+\sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\, \delta Y_{s,t}^{\kappa} + \sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa (\Gamma_s^{\mathbf{Y}})_\lambda u_s + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa} u_s\Big)\, \mathbb{Y}_{s,t}^{\kappa\lambda} + \mu_s(u_{s,t}^{\natural}),$$

where $|\mu_s(u_{s,t}^{\natural})| \lesssim |t-s|^{3\alpha}$ since $\sup_{t\in[0,\tau]} |u_t|_{\mathcal{C}_b^3} < +\infty$.

Proof of Theorem 13

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \le s \le t \le T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Second term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_s) - \mu_s(u_s)$:

$$+(\mu_t - \mu_s)(u_s) = \mu_t(u_s) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr$$

$$+\sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\, \delta Y_{s,t}^{\kappa} + \sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_t^{\mathbf{Y}})_\kappa (\Gamma_t^{\mathbf{Y}})_\lambda u_t + (\Gamma_t^{\mathbf{Y}})'_{\lambda\kappa} u_t + P_{s,t}\Big)\, \mathbb{Y}_{s,t}^{\kappa\lambda} + \mu_s(u_{s,t}^{\natural}),$$

where $|P_{s,t}| \lesssim |t-s|^\alpha$ (u solves Equation (5): Condtition (ii)).

Proof of Theorem 13

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \leq s \leq t \leq T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Second term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_s) - \mu_s(u_s)$:

$$+(\mu_t - \mu_s)(u_s) = \mu_t(u_s) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr$$

$$+\sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\, \delta Y_{s,t}^{\kappa} + \sum_{\kappa,\lambda=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_t^{\mathbf{Y}})_\kappa (\Gamma_t^{\mathbf{Y}})_\lambda u_t + (\Gamma_t^{\mathbf{Y}})'_{\lambda\kappa} u_t\Big)\, \mathbb{Y}_{s,t}^{\kappa\lambda} + \mu_s(u_{s,t}^{\natural}),$$

where $|\mu_s(u_{s,t}^{\natural})| \lesssim |t-s|^{3\alpha}$ since $|\mu_s(P_{s,t})\mathbb{Y}_{s,t}^{\kappa\lambda}| \lesssim |t-s|^{3\alpha}$.

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \leq s \leq t \leq T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Third term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_t - u_s) - \mu_s(u_t - u_s)$:

$$+(\mu_t - \mu_s)(u_t - u_s) = \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa (u_t - u_s)\Big)\, \delta Y_{s,t}^{\kappa} + Q_{s,t}$$

$$= \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa u_t - (\Gamma_t^{\mathbf{Y}})_\kappa u_t\Big)\, \delta Y_{s,t}^{\kappa} \; + \; \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_t^{\mathbf{Y}})_\kappa u_t - (\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\, \delta Y_{s,t}^{\kappa} \; + \; Q_{s,t},$$

where $|Q_{s,t}| \lesssim |t-s|^{3\alpha}$ uniformly in s, t since $u \in \mathcal{C}^{\alpha}([0, \tau]); C_b^2)$.

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \le s \le t \le T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Third term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_t - u_s) - \mu_s(u_t - u_s)$:

$$+(\mu_t - \mu_s)(u_t - u_s) = \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa(u_t - u_s)\Big)\,\delta Y_{s,t}^\kappa + Q_{s,t}$$

$$= \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa u_t - (\Gamma_t^{\mathbf{Y}})_\kappa u_t\Big)\,\delta Y_{s,t}^\kappa \;+\; \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_t^{\mathbf{Y}})_\kappa u_t - (\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\,\delta Y_{s,t}^\kappa \;+\; Q_{s,t},$$

where $|Q_{s,t}| \lesssim |t-s|^{3\alpha}$ uniformly in s, t since $u \in \mathcal{C}^\alpha([0,\tau]); C_b^2)$.

Proof of Theorem 13

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \le s \le t \le T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Third term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_t - u_s) - \mu_s(u_t - u_s)$:

$$+(\mu_t - \mu_s)(u_t - u_s) = \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa (u_t - u_s)\Big)\,\delta Y_{s,t}^{\kappa} + Q_{s,t}$$

$$= \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\left(-\sum_{\lambda=1}^{d_Y} (\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t \delta Y_{s,t}^{\lambda} + R_{s,t}\right)\delta Y_{s,t}^{\kappa} \;+\; \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_t^{\mathbf{Y}})_\kappa u_t - (\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\,\delta Y_{s,t}^{\kappa} \;+\; Q_{s,t},$$

where $\sup_{x \in \mathbb{R}^{d_X}} |R_{s,t}(x)| \lesssim |t-s|^{2\alpha}$ uniformly in s, t thanks to Assumption U.

For $\mu \in \{\mu^1, \mu^2\}$ and for any $0 \leq s \leq t \leq T$ we can write

$$\mu_t(u_t) - \mu_s(u_s) = -\mu_s(u_s - u_t) + (\mu_t - \mu_s)(u_s) + (\mu_t - \mu_s)(u_t - u_s).$$

Third term: μ solves the Zakai Equation (4) so we can use the Davie type expansion for $\mu_t(u_t - u_s) - \mu_s(u_t - u_s)$:

$$+(\mu_t - \mu_s)(u_t - u_s) = \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa (u_t - u_s)\Big)\, \delta Y_{s,t}^{\kappa} + Q_{s,t}$$

$$= \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\left(\sum_{\lambda=1}^{d_Y} (\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t \delta Y_{s,t}^{\lambda}\right) \delta Y_{s,t}^{\kappa} \; + \; \sum_{\kappa=1}^{d_{\mathbf{Y}}} \mu_s\Big((\Gamma_t^{\mathbf{Y}})_\kappa u_t - (\Gamma_s^{\mathbf{Y}})_\kappa u_s\Big)\, \delta Y_{s,t}^{\kappa} \; + \; Q_{s,t},$$

where $|Q_{s,t}| \lesssim |t-s|^{3\alpha}$ since $|R_{s,t} \delta Y_{s,t}^{\kappa}| \lesssim |t-s|^{3\alpha}$.

Therefore, by adding the identities and cancelling the matching terms, we end up with

$$\mu_t(u_t) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr \;-\; \mu_s\Big(\int_s^t A_r^{\mathbf{Y}} u_r\, dr\Big) - \mu_s\left(\sum_{\kappa,\lambda=1}^{d_Y} \int_s^t (\Gamma_r^{\mathbf{Y}})'_{\kappa\lambda} u_r\, [\dot{\mathbf{Y}}_r]^{\kappa\lambda}\, dr\right)$$

$$+ \sum_{\kappa,\lambda=1}^{d_X} \mu_s\Big((\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t\Big)\, (\mathbb{Y}_{s,t}^{\kappa\lambda} + \mathbb{Y}_{s,t}^{\lambda\kappa} - \delta Y_{s,t}^{\lambda}\, \delta Y_{s,t}^{\kappa}) \;+\; S_{s,t},$$

with $\quad |S_{s,t}| \lesssim |t-s|^{3\alpha}$.

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$$+ \sum_{\kappa,\lambda=1}^{d_X} \mu_s\Big((\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t\Big)\, (\mathbb{Y}_{s,t}^{\kappa\lambda} + \mathbb{Y}_{s,t}^{\lambda\kappa} - \delta Y_{s,t}^{\lambda}\, \delta Y_{s,t}^{\kappa}) \; + \; S_{s,t},$$

with $\quad |S_{s,t}| \lesssim |t-s|^{3\alpha}.$

Using $x \mapsto A_t^{\mathbf{Y}} u_t(x)$ is Lipschitz and bounded uniformly in t and Hypothesis 6:

$$\left|\int_s^t \mu_r\Big(A_r^{\mathbf{Y}} u_s\Big)\, dr - \mu_s\left(\int_s^t A_r^{\mathbf{Y}} u_r\, dr\right)\right| \leq \left|\int_s^t \mu_r\Big(A_r^{\mathbf{Y}}(u_s - u_r)\Big)\, dr\right| + \left|\int_s^t (\mu_r - \mu_s)(A_r^{\mathbf{Y}} u_r)\, dr\right|$$

$$\lesssim |u|_{C^\alpha} \sup_{r\in[0,\tau]} |\mu_r(1)|\, |t-s|^{1+\alpha} + |t-s|^{1+\alpha}.$$

Therefore, by adding the identities and cancelling the matching terms, we end up with

$$\mu_t(u_t) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr \;-\; \mu_s\Big(\int_s^t A_r^{\mathbf{Y}} u_r\, dr\Big) - \mu_s\left(\sum_{\kappa,\lambda=1}^{d_Y} \int_s^t (\Gamma_r^{\mathbf{Y}})'_{\kappa\lambda} u_r\, [\dot{\mathbf{Y}}_r]^{\kappa\lambda}\, dr\right)$$

$$+ \sum_{\kappa,\lambda=1}^{d_X} \mu_s\Big((\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t\Big)\, (\mathbb{Y}_{s,t}^{\kappa\lambda} + \mathbb{Y}_{s,t}^{\lambda\kappa} - \delta Y_{s,t}^{\lambda}\, \delta Y_{s,t}^{\kappa}) \;+\; S_{s,t},$$

with $\quad |S_{s,t}| \lesssim |t-s|^{3\alpha}.$

Therefore, by adding the identities and cancelling the matching terms, we end up with

$$\mu_t(u_t) - \mu_s(u_s) = \int_s^t \mu_r(A_r^{\mathbf{Y}} u_s)\, dr \,-\, \mu_s\Big(\int_s^t A_r^{\mathbf{Y}} u_r\, dr\Big) - \mu_s\left(\sum_{\kappa,\lambda=1}^{d_Y} \int_s^t (\Gamma_r^{\mathbf{Y}})'_{\kappa\lambda} u_r\, [\dot{\mathbf{Y}}_r]^{\kappa\lambda}\, dr\right)$$

$$+ \sum_{\kappa,\lambda=1}^{d_X} \mu_s\Big((\Gamma_t^{\mathbf{Y}})'_{\kappa\lambda} u_t\Big)\, (\mathbb{Y}_{s,t}^{\kappa\lambda} + \mathbb{Y}_{s,t}^{\lambda\kappa} - \delta Y_{s,t}^{\lambda}\, \delta Y_{s,t}^{\kappa}) \,+\, S_{s,t},$$

with $\quad |S_{s,t}| \lesssim |t-s|^{3\alpha}.$

Remember that $\mathbb{Y}_{s,t}^{\kappa\lambda} + \mathbb{Y}_{s,t}^{\lambda\kappa} - \delta Y_{s,t}^{\kappa}\, \delta Y_{s,t}^{\lambda} = -\delta[\mathbf{Y}]_{s,t}^{\kappa\lambda} = -\int_s^t [\dot{\mathbf{Y}}_r]^{\kappa\lambda}\, dr$ and under Assumption U $\sup_{x\in\mathbb{R}^{d_X}} \left|(\Gamma_t^{\mathbb{Y}})'_{\kappa\lambda} u_t(x) - (\Gamma_s^{\mathbb{Y}})'_{\kappa\lambda} u_s(x)\right| \lesssim |t-s|^{\alpha}$:

$$\left|\mu_s\left(\sum_{\kappa,\lambda=1}^{d_Y} \int_s^t (\Gamma_r^{\mathbb{Y}})'_{\kappa\lambda} u_r\, [\dot{\mathbf{Y}}_r]^{\lambda}\, dr\right) - \sum_{\kappa,\lambda=1}^{d_X} \mu_s\Big((\Gamma_t^{\mathbb{Y}})'_{\kappa\lambda} u_t\Big) \int_s^t [\dot{\mathbf{Y}}_r]^{\kappa\lambda}\, dr\right| \lesssim |t-s|^{1+\alpha}$$

$\square$

Rough stochastic filtering

Rough Kushner-Stratonovich equation

Theorem 14 ([CP24], Theorem 3.11 and Theorem 3.17)

Under Assumption CP-E (in the paper), the measure-valued process $\varsigma_t(\varphi) = \mathbb{E}^{\circ}\left[\varphi(X_t) \mid \mathcal{F}_t^Y\right]$. *Kushner–Stratonovich equation, for all* $\varphi \in C_b^2(\mathbb{R}^{d_x}; \mathbb{R})$:

$$
\begin{aligned}
\varsigma_t(\varphi) = \varsigma_0(\varphi) + \int_0^t \varsigma_r(A_r\varphi)\, dr \\
+ \int_0^t \left(\varsigma_r(\nabla\varphi\, f) + \varphi h - \varsigma_r(\varphi)\varsigma_r(h)\right)\left(dY_r - \frac{d\langle Y, Y\rangle_r}{dr}\,\varsigma_r(h^\top)\, dr\right)
\end{aligned}
$$

Moreover, the uniqueness of the solution of the above equation is equivalent to that of the stochastic Zakai equation.

In the same spirit of the rough Zakai equation, we want to give meaning to an equation of the form

$$\begin{aligned}\varsigma_t^{\mathbf{Y}}(\varphi) &= \varsigma_0^{\mathbf{Y}}(\varphi) + \int_0^t \varsigma_r^{\mathbf{Y}}(A_r^{\mathbf{Y}}\varphi)\,dr \\ &\quad + \int_0^t \Big(\varsigma_r^{\mathbf{Y}}(\Gamma_r^{\mathbf{Y}}\varphi) - \varsigma_r^{\mathbf{Y}}(\varphi)\,\varsigma_r^{\mathbf{Y}}(h(r,\cdot;\mathbf{Y}))\Big)\Big(d\mathbf{Y}_r - [\dot{\mathbf{Y}}]_r\,\varsigma_r(h(r,\cdot;\mathbf{Y}))^T\,dr\Big),\end{aligned} \tag{7}$$

where we recall that $\varsigma_t^{\mathbf{Y}} := \frac{\mu_t^{\mathbf{Y}}(\varphi)}{\mu_t^{\mathbf{Y}}(1)}$.

For sake of a more compact notation, for any $\varsigma : [0, T] \to \mathcal{M}_{\mathcal{F}}(\mathbb{R}^{d_X})$ and for any sufficiently regular test function $\varphi : \mathbb{R}^{d_x} \to \mathbb{R}$, let us define $\Phi_t^{\varsigma,\varphi} = ((\Phi_t^{\varsigma,\varphi})_1, \dots, (\Phi_t^{\varsigma,\varphi})_{d_Y})$ as

$$(\Phi_t^{\varsigma,\varphi})_\kappa := \varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa \varphi) - \varsigma_t(\varphi)\varsigma_t(h_\kappa(t, \cdot; \mathbf{Y})) \qquad \kappa = 1, \dots, d_y.$$

Rough Kushner–Stratonovich equation (definition)

For sake of a more compact notation, for any $\varsigma : [0, T] \to \mathcal{M}_{\mathcal{F}}(\mathbb{R}^{d_X})$ and for any sufficiently regular test function $\varphi : \mathbb{R}^{d_x} \to \mathbb{R}$, let us define $\Phi_t^{\varsigma,\varphi} = ((\Phi_t^{\varsigma,\varphi})_1, \dots, (\Phi_t^{\varsigma,\varphi})_{d_Y})$ as

$$(\Phi_t^{\varsigma,\varphi})_\kappa := \varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa \varphi) - \varsigma_t(\varphi)\varsigma_t(h_\kappa(t, \cdot; \mathbf{Y})) \qquad \kappa = 1, \dots, d_y.$$

Now we would like to give a definition for the Gubinelli derivative of $(\Phi_t^{\varsigma,\varphi})_\kappa$:

$$(\Phi_t^{\varsigma,\varphi})' := \begin{pmatrix} (\Phi_t^{\varsigma,\varphi})'_{11} & \cdots & (\Phi_t^{\varsigma,\varphi})'_{1d_y} \\ \vdots & \ddots & \vdots \\ (\Phi_t^{\varsigma,\varphi})'_{d_y 1} & \cdots & (\Phi_t^{\varsigma,\varphi})'_{d_y d_y} \end{pmatrix}$$

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Thanks to the product rule for Gubinelli derivatives:

$$(\Phi_t^{\varsigma,\varphi})'_{\kappa\lambda} := \varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa \varphi)'_\lambda - \varsigma_t(\varphi)'_\lambda \varsigma_t(h_\kappa(t, \cdot; \mathbf{Y})) - \varsigma_t(\varphi)\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa 1)'_\lambda, \qquad \kappa, \lambda = 1, \ldots, d_y,$$

where we used the fact that $h_\kappa(t, x; \mathbf{Y}) = (\Gamma_t^{\mathbf{Y}})_\kappa 1(x)$.

To understand what the appropriate definitions to give for $\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa\varphi)'_\lambda$ and $\varsigma_t(\varphi)'_\lambda$ should be, we can consider these formal identities:

$$\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa\varphi) - \varsigma_s((\Gamma_s^{\mathbf{Y}})_\kappa\varphi) \approx \sum_{\eta=1}^{d_Y} \left[\varsigma_s((\Gamma_s^{\mathbf{Y}})_\eta(\Gamma_s^{\mathbf{Y}})_\kappa\varphi) + (\Gamma_s^{\mathbf{Y}})'_{\kappa\eta}\varphi - \varsigma_s((\Gamma_s^{\mathbf{Y}})_\kappa\varphi)\varsigma_s(h_\eta(s,\cdot;\mathbf{Y}))\right]\delta Y^\eta_{s,t},$$

$$\varsigma_t(\varphi) - \varsigma_s(\varphi) \approx \sum_{\eta=1}^{d_Y}[\varsigma_s((\Gamma_s^{\mathbf{Y}})_\eta\varphi) - \varsigma_s(\varphi)\varsigma_s(h_\eta(s,\cdot;\mathbf{Y}))]\delta Y^\eta_{s,t}.$$

Hence, it is justified to define

$$\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa\varphi)'_\eta := \varsigma_s((\Gamma_s^{\mathbf{Y}})_\eta(\Gamma_s^{\mathbf{Y}})_\kappa\varphi) + (\Gamma_s^{\mathbf{Y}})'_{\kappa\eta}\varphi - \varsigma_s((\Gamma_s^{\mathbf{Y}})_\kappa\varphi)\varsigma_s(h_\eta(s,\cdot;\mathbf{Y})),$$

$$\varsigma_t(\varphi)'_\eta := \varsigma_s((\Gamma_s^{\mathbf{Y}})_\eta\varphi) - \varsigma_s(\varphi)\varsigma_s(h_\eta(s,\cdot;\mathbf{Y})).$$

A measure-valued solution to Equation (7) is a weakly continuous curve

$$\varsigma : [0, T] \to \mathcal{M}_{\mathcal{F}}(\mathbb{R}^{d_x})$$

such that, for any test function $\varphi \in \mathcal{C}_b^3(\mathbb{R}^{d_x}; \mathbb{R})$, the following hold:

(i) the maps $t \mapsto \varsigma_t(A_t^{\mathbf{Y}}\varphi)$ and $t \mapsto \sum_{\kappa=1}^{d_y} \Big(\varsigma_t\big((\Gamma_t^{\mathbf{Y}})_\kappa\varphi\big) - \varsigma_t(\varphi)\varsigma_t(h_\kappa(t, \cdot; \mathbf{Y}))\Big)[\dot{\mathbf{Y}}]_t^\kappa$ belong to $L^1([0, T], dt)$.

(ii) There exists $C = C_\varphi > 0$ uniform over bounded sets of $\varphi \in \mathcal{C}_b^3(\mathbb{R}^{d_x}; \mathbb{R})$ such that, for any $0 \le s \le t \le T$ and for any $\kappa, \lambda = 1, \ldots, d_y$:

$$\big|(\Phi_t^{\varsigma,\varphi})'_{\kappa\lambda} - (\Phi_s^{\varsigma,\varphi})'_{\kappa\lambda}\big| \le C_\varphi |t-s|^\alpha,$$

and

$$\Big|(\Phi_t^{\varsigma,\varphi})_\kappa - (\Phi_s^{\varsigma,\varphi})_\kappa - \sum_{\eta=1}^{d_y} (\Phi_s^{\varsigma,\varphi})'_{\kappa\eta}\delta Y_{s,t}^\eta\Big| \le C_\varphi |t-s|^{2\alpha}.$$

(iii) For any $0 \leq s \leq t \leq T$, it holds the Davie-type expansion

$$\varsigma_t(\varphi) - \varsigma_s(\varphi) = \int_s^t \varsigma_r(A_r^{\mathbf{Y}}\varphi)\, dr - \sum_{\kappa,\lambda=1}^{d_Y} (\Phi_r^{\varsigma,\varphi})_\kappa [\dot{\mathbf{Y}}]_r^{\kappa,\lambda} \varsigma_r(h_\lambda(r,\cdot;\mathbf{Y}))\, dr$$
$$+ \sum_{\kappa=1}^{d_Y} (\Phi_s^{\varsigma,\varphi})_\kappa \delta Y_{s,t}^\kappa \quad + \sum_{\kappa,\lambda=1}^{d_Y} (\Phi_s^{\varsigma,\varphi})'_{\kappa\lambda} \mathbb{Y}_{s,t}^{\kappa\lambda} + \mu_{s,t}^{\varphi,\varsigma},$$

where

$$\mu_{s,t}^{\varphi,\varsigma} = o(|t-s|) \quad \text{as } |t-s| \to 0.$$

If the initial value $\varsigma_0 = \nu \in \mathcal{M}_{\mathcal{F}}(\mathbb{R}^{d_x})$ is specified, we say that ς is a solution starting from ν.

Existence for rough Kushner-Stratonovich equation

Existence for rough Kushner–Stratonovich follows from existence for rough Zakai, while uniqueness of the former is equivalent to uniqueness of the latter.

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Theorem 15

Let $\mu : [0, T] \to \mathcal{M}(\mathbb{R}^{d_x})$ be a measure-valued solution to the rough Zakai equation (4) such that $\mu_t(1) > 0$ for any $t \in [0, T]$. Define the measure-valued path $\varsigma : [0, T] \to \mathcal{M}(\mathbb{R}^{d_x})$ via

$$\varsigma_t(\varphi) := \frac{\mu_t(\varphi)}{\mu_t(1)}$$

for any $\varphi : \mathbb{R}^{d_X} \to \mathbb{R}$ measurable and bounded. Then, the path ς solves the rough Kushner-Stratonovich PDE:

$$\begin{aligned}\varsigma_t(\varphi) = \varsigma_0(\varphi) &+ \int_0^t \varsigma_r(A_r^{\mathbf{Y}}\varphi)\, dr \\ &+ \int_0^t \big(\varsigma_r(\Gamma_r^{\mathbf{Y}}\varphi) - \varsigma_r(\varphi)\varsigma_r(h(r,\cdot;\mathbf{Y}))\big)\big(d\mathbf{Y}_r - [\dot{\mathbf{Y}}]_r\varsigma_r(h(r,\cdot;\mathbf{Y}))^\top dr\big)\end{aligned}$$

- ς **is weakly continuous.** This follows directly from the definition of ς and from the assumed weak continuity of μ.

- **Measurability and integrability**
  - The map $[0,T] \ni t \mapsto \varsigma_t(A_t^{\mathbf{Y}}\varphi) \in \mathbb{R}$ is measurable (because $t \mapsto \mu_t(A_t^{\mathbf{Y}}\varphi)$ is measurable and $t \mapsto \mu_t(1)$ is continuous) and bounded away from 0. Moreover,

$$\int_0^T \left|\varsigma_t(A_t^{\mathbf{Y}}\varphi)\right| dt \leq \frac{1}{\min_{0\leq t\leq T}\mu_t(1)} \int_0^T \left|\mu_t(A_t^{\mathbf{Y}}\varphi)\right| dt < +\infty.$$

  - The map $t \mapsto \sum_{\kappa,\lambda=1}^{d_Y} \big(\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa\varphi) - \varsigma_t(\varphi)\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa 1)\big)[\dot{\mathbf{Y}}]_t^\kappa\, \varsigma_t((\Gamma_t^{\mathbf{Y}})_\lambda 1)$.

- **Condition** (ii)
  Straightforward from the tow formal identities used to define $\varsigma_t((\Gamma_t^{\mathbf{Y}})_\kappa\varphi)'_\eta$ and $\varsigma_t(\varphi)'_\eta$.

- **Condition** (iii): we recall

Theorem 16 ([FH20], Theorem 7.7)

Let $F : V \to W$ be a C^3 map. Let $\mathbf{Y} = (Y, \mathbb{Y}) \in \mathcal{C}^\alpha$ be a rough path, and let $(X, X') \in \mathcal{D}_Y^{2\alpha}$ be a controlled rough path satisfying

$$X_t = X_0 + \int_0^t X'_s \, d\mathbf{Y}_s + \Gamma_t,$$

for some rough controlled path $(X', X'') \in \mathcal{D}_Y^{2\alpha}$ and $\Gamma \in C^{2\alpha}$. Then, $(DF(X)X', D^2F(X)(X', X') + DF(X)X'') \in \mathcal{D}_Y^{2\alpha}$ and the following Itô-type formula holds:

$$\boxed{F(X_t) = F(X_0) + \int_0^t DF(X_s)X'_s \, d\mathbf{Y}_s + \int_0^t DF(X_s)\, d\Gamma_s + \frac{1}{2}\int_0^t D^2F(X_s)(X'_s, X'_s)\, d[\mathbf{Y}]_s.}$$

Since $A_t^{\mathbf{Y}}1 = 0$, using the Davie-type expansion for the tough Zakai equation of $\mu(1)$:

$$\mu_t(1) - \mu_s(1) = \mu_s\big((\Gamma_s^{\mathbf{Y}})_\kappa\, 1\big)\,\delta Y_{s,t}^\kappa \;+\; \mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda 1 \;+\; (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa}1\Big)\,\mathbb{Y}_{s,t}^{\kappa\lambda} \;+\; \mu_{s,t}^{1,\natural}.$$

So, to compute the Davie-type expansion for $1/\mu(1)$, we can use the rough Itô formula where:

$$X_t = \mu_t(1), \quad X_t' = \mu_t\big((\Gamma_t^{\mathbf{Y}})1\big), \quad X_t'' = \mu_t\Big((\Gamma_t^{\mathbf{Y}})(\Gamma_t^{\mathbf{Y}})1 \;+\; (\Gamma_t^{\mathbf{Y}})'1\Big), \quad F(x) = \frac{1}{x}, \quad \Gamma_t = 0.$$

In particular:

$$\int_0^t DF(X_s)X_s'\, d\mathbf{Y}_s \approx -\frac{\mu_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)}{\mu_s(1)^2}\delta Y_{s,t}^\kappa + \left(\frac{2\mu_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)\mu_s\big((\Gamma_s^{\mathbf{Y}})_\lambda 1\big)}{\mu_s(1)^3} - \frac{\mu_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda 1 + (\Gamma_s^{\mathbf{Y}})'_{\kappa,\lambda}1\Big)}{\mu_s(1)^2}\right)\mathbb{Y}_{s,t}^{\kappa\lambda}$$

$$= -\frac{\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)}{\mu_s(1)}\delta Y_{s,t}^\kappa + \left(\frac{2\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\lambda 1\big) - \varsigma_s\Big((\Gamma_s^{\mathbf{Y}})_\kappa(\Gamma_s^{\mathbf{Y}})_\lambda\, 1 + (\Gamma_s^{\mathbf{Y}})'_{\kappa,\lambda}1\Big)}{\mu_s(1)}\right)\mathbb{Y}_{s,t}^{\kappa\lambda};$$

$$\frac{1}{2}\int_0^t D^2F(X_s)(X_s', X_s')\, d[\mathbf{Y}]_s \approx \int_s^t \frac{\mu_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)\mu_s\big((\Gamma_s^{\mathbf{Y}})_\lambda 1\big)}{\mu_r^3(1)}\,[\dot{\mathbf{Y}}]_s^{\kappa\lambda}\, ds = \int_s^t \frac{\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\lambda 1\big)}{\mu_r(1)}\,[\dot{\mathbf{Y}}]_s^{\kappa\lambda}\, ds.$$

Collecting all terms, we obtain:

$$\frac{1}{\mu_t(1)} - \frac{1}{\mu_s(1)} = \int_s^t \varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)\, \varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\lambda 1\big)\, ([\dot{\mathbf{Y}}]_r)^{\kappa\lambda}\, \frac{1}{\mu_r(1)}\, dr - \frac{\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big)}{\mu_s(1)}\, \delta Y^\kappa_{s,t}$$

$$+ \Big(-\varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa (\Gamma_s^{\mathbf{Y}})_\lambda 1\big) + (\Gamma_s^{\mathbf{Y}})'_{\lambda\kappa} 1 + 2\, \varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\kappa 1\big) \varsigma_s\big((\Gamma_s^{\mathbf{Y}})_\lambda 1\big)\Big) \frac{1}{\mu_s(1)}\, \mathbb{Y}^{\kappa\lambda}_{s,t} + P_{s,t}$$

where $|P_{s,t}| \lesssim |t-s|^{3\alpha}$. Using the product formula for controlled paths $(x \cdot \frac{1}{y})$ shows immediately that ς satisfies Condition (iii).

$\square$

Condition (CO):
We say that a solution ς to the rough Kushner-Stratonovich equation (7) satisfies the condition (CO) if

$$(\Psi^{\varsigma}, (\Psi^{\varsigma})') \in \mathcal{D}_Y^{2\alpha}\big([0,T], \mathbb{R}^{1\times d_Y}\big) \tag{48}$$

is a (deterministic) controlled rough path, where

$$(\Psi_t^{\varsigma})_{\kappa} := \varsigma_t\big(h_{\kappa}(t,\cdot\,;\mathbf{Y})\big),$$

$$(\Psi_t^{\varsigma})'_{\kappa\lambda} := \varsigma_t\big((\Gamma_t^{\mathbf{Y}})_{\lambda} h_{\kappa}(t,\cdot\,;\mathbf{Y}) + h'_{\kappa\lambda}(t,\cdot\,;\mathbf{Y})\big) - \varsigma_t\big(h_{\kappa}(t,\cdot\,;\mathbf{Y})\big)\,\varsigma_t\big(h_{\lambda}(t,\cdot\,;\mathbf{Y})\big),$$

for $t \in [0,T]$ and $\kappa, \lambda = 1, \ldots, d_Y$.

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is a (deterministic) controlled rough path, where

$$(\Psi^{\varsigma}_t)_{\kappa} := \varsigma_t\big(h_{\kappa}(t, \cdot\,; \mathbf{Y})\big),$$

$$(\Psi^{\varsigma}_t)'_{\kappa\lambda} := \varsigma_t\big((\Gamma^{\mathbf{Y}}_t)_{\lambda} h_{\kappa}(t, \cdot\,; \mathbf{Y}) + h'_{\kappa\lambda}(t, \cdot\,; \mathbf{Y})\big) - \varsigma_t\big(h_{\kappa}(t, \cdot\,; \mathbf{Y})\big)\, \varsigma_t\big(h_{\lambda}(t, \cdot\,; \mathbf{Y})\big),$$

for $t \in [0, T]$ and $\kappa, \lambda = 1, \dots, d_Y$.

Remkark: (CO) is satisfied if $[t \mapsto (h(t, \cdot; \mathbf{Y})), (h'(t, \cdot; \mathbf{Y}))] \in \mathcal{D}_Y^{2\alpha}\mathcal{C}_b^{\beta}$. In particular, Assumption E implies condition (CO).

First of all, we need a preliminary lemma:

Lemma 17

Let ς be a measure-valued solution to the rough Kushner-Stratonovich equation (7) satisfying condition (CO). Then, the following linear RDE is well-posed:

$$dZ_t^{\varsigma} = \varsigma_t\big(h(t,\,\cdot\,;\mathbf{Y})\big)\, Z_t^{\varsigma}\; d\mathbf{Y}_t. \tag{8}$$

In particular, its unique solution Z^{ς} is strictly positive if $Z_0^{\varsigma} > 0$. Moreover, if $\varsigma_0(1) = 1$ the following hold true:

i. *every ς_t is a probability measure on $\mathbb{R}^{d_Y}$;*
ii. *$Z^{\varsigma}\varsigma$ is a solution to the rough Zakai equation (4) starting from $Z_0^{\varsigma}\varsigma_0$.*

Theorem 18

Let $\mu_0 = \varsigma_0$ be probability measures on $\mathbb{R}^{d_X}$. Then the following are equivalent:

1. *there exists at most one measure-valued solution μ to the rough Zakai equation* (4) *starting from μ_0 and such that $\mu_t(1) > 0$ for any $t \in [0, T]$;*
2. *there exists at most one measure-valued solution ς of the rough Kushner–Stratonovich* (7) *starting from ς_0 and such that condition (CO) holds.*

$(1) \Longrightarrow (2)$: Let ς^1 and ς^2 defined as in (2). By Lemma 17, we can consider Z^{ς^1} and Z^{ς^2} as the solutions to the previous linear RDE starting from $Z_0^{\varsigma^1} = Z_0^{\varsigma^2} = 1$. Thanks to the second part of the lemma ($\varsigma_0(1) = 1$), both $Z_t^{\varsigma^1}\varsigma_t^1$ and $Z_t^{\varsigma^2}\varsigma_t^2$ are solutions to the rough Zakai equation starting from ς_0 and ς_t^1 and ς_t^2 are probability measures for any $t \in [0, T]$. From this last property:

$$Z_t^{\varsigma^i}\varsigma_t^i(1) = Z_t^{\varsigma^i} > 0 \qquad \text{for any } t \in [0, T].$$

We are under the right hypothesis in (1), hence $Z_t^{\varsigma^1}\varsigma_t^1 = Z_t^{\varsigma^2}\varsigma_t^2$. In particular

$$Z_t^{\varsigma^1} = Z_t^{\varsigma^1}\varsigma_t^1(1) = Z_t^{\varsigma^2}\varsigma_t^2(1) = Z_t^{\varsigma^2} \qquad \text{for any } t \in [0, T].$$

Recalling that $Z_t^i(1) > 0$ for any $t \in [0, T]$, we have $\varsigma^1 = \varsigma^2$.

$(2) \implies (1)$: Let μ^1 and μ^2 defined as in (1). Setting $\varsigma^i := \frac{\mu^i}{\mu^i(1)}$, $i = 1, 2$, we have that ς^1 and ς^2 both solve the rough Kushner-Stratonoivch equation and satisfy condition (CO); hence, by uniqueness, $\varsigma^1 = \varsigma^2 = \varsigma$.

Recalling that $(\mu^i_t(1))' = \mu_t(\Gamma^{\mathbf{Y}}_t 1)$, we have:

$$\mu^i_t(1) = 1 + \int_0^t \frac{\mu^i_r(\Gamma^{\mathbf{Y}}_r 1)}{\mu^i_r(1)} \mu^i_r(1)\, dY_r, \qquad t \in [0, T], \; i = 1, 2.$$

So, $Z^i_t = \mu^i_t(1)$ solves the linear equation in Lemma 17 of Theorem 3.19 with $Z^i_0 = 1$, for $i = 1, 2$. Hence, $Z^1 = Z^2$. We conclude that

$$\mu^1_t = \mu^1_t(1)\, \varsigma^1_t = \mu^2_t(1)\, \varsigma^2_t = \mu^2_t \qquad \text{for any } t \in [0, T].$$

$\square$

Outline

Setting

- Assume it holds:

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P}) = (\Omega', \mathcal{F}', (\mathcal{F}'_t)_t, \mathbb{P}') \otimes (\Omega'', \mathcal{F}'', (\mathcal{F}''_t)_t, \mathbb{P}'')$$

  where $(\Omega', \mathcal{F}', (\mathcal{F}'_t)_t, \mathbb{P}')$ is a complete probability space supporting a d_B-dimensional Brownian motion B, similar for $(\Omega'', \mathcal{F}'', (\mathcal{F}''_t)_t, \mathbb{P}'')$ with a d_W-dimensional Brownian motion W.

- We take $\mathcal{C}_T := \mathcal{C}_T^{0,\alpha,1}$ with $\alpha \in (1/3, 1/2)$ and equip it with the Borel sigma algebra $\mathcal{C}_T$.

- **Assumption CME.** Assume Assumption E holds and:

  - *Dependence on* $\mathbf{Y}$. The maps $(b, \sigma, (f, f'), (h, h'))$ as functions of $(t, x, \mathbf{Y})$ are $\mathcal{B}_{[0,t]} \otimes \mathcal{B}(\mathbb{R}^d_x) \otimes \mathcal{C}_T / \mathcal{B}(\mathbb{R}^{d'})$-measurable and uniformly bounded, with appropriate choices of d';

  - *Non-anticipative structure.* All coefficient functions above are causal in Y, i.e. for any $\phi := b, \sigma, f, f', h, h'$, $\phi(t, x, Y) = \phi(t, x, Y^t)$, where $Y^t_s := Y_{t \wedge s}$.

- **Assumption OWP.** Assume strong well-posedness of

$$Y_t(\omega'') = \int_0^t k(r, Y_r)\, dW_r(\omega'').$$

(Of course, the usual Lipschitz condition on k as contained in Assumption CP-E is sufficient.)

By assumption, Y is a local martingale, hence admits a rough path lift,

$$\mathbf{Y}^{\mathrm{Ito}}(\omega'') := (Y, \mathbb{Y}^{\mathrm{Ito}})(\omega'')$$

be the Itô (martingale) rough path, with

$$\mathbb{Y}^{\mathrm{Ito}}_{s,t}(\omega'') := \left(\int_s^t \delta Y_{s,r} \otimes dY_r\right)(\omega''), \qquad \forall\, 0 \le s \le t \le T.$$

As Y is driven by the Brownian motion and hence $\mathcal{F}''_t$-measurable, and as the second level is non–anticipative as well, it follows that $\mathcal{F}^{\mathbf{Y}^{\mathrm{Ito}}}_t \subset \mathcal{F}''_t$.

Connection rough and stochastic filtering

Stochastic vs rough Kallianpur-Striebel formula

Measurability and Non-Anticipativity of RSDE Solutions

Recall that, by definition, for any bounded Borel measurable function φ on $\mathbb{R}^{d_X}$,

$$\mu_t^{\mathbf{Y}}(\varphi) = \mathbb{E}\big[\varphi(X_t^{\mathbf{Y}}) Z_t^{\mathbf{Y}}\big], \qquad \varsigma_t^{\mathbf{Y}}(\varphi) = \frac{\mu_t^{\mathbf{Y}}(\varphi)}{\mu_t^{\mathbf{Y}}(1)},$$

where $(X^{\mathbf{Y}}, Z^{\mathbf{Y}})$ is the solution of rough SDEs (2) and (4) with $(\Omega, \mathcal{F}, \mathbb{P})$ replaced by $(\Omega', \mathcal{F}', \mathbb{P}')$.

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where $(X^{\mathbf{Y}}, Z^{\mathbf{Y}})$ is the solution of rough SDEs (2) and (4) with $(\Omega, \mathcal{F}, \mathbb{P})$ replaced by $(\Omega', \mathcal{F}', \mathbb{P}')$.

Lemma 19 ([FLZ25], Theorem 3.4.)

Under Assumption CME,

(i) *there is a $\mathcal{F}'_t \otimes \mathcal{C}_t$-progressively measurable version of $(X^{\mathbf{Y}}, Z^{\mathbf{Y}})$;*

(ii) *for any $t \geq 0$, we also get a $\mathcal{C}_T \otimes \mathcal{B}_{[0,t]}$-measurable mapping $(\mathbf{Y}, r) \mapsto \mu_r^{\mathbf{Y}} \in \mathcal{M}_F$. Similar for $\varsigma_t^{\mathbf{Y}}$ as a functional with values in $\mathcal{P}$.*

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By measurability of $\omega'' \mapsto \mathbf{Y}^{\mathrm{Ito}}(\omega'')$ together with (i) above, it is clear that

$$(\bar{X}, \bar{Z})_t(\omega', \omega'') := (X^{\mathbf{Y}}, Z^{\mathbf{Y}})_t(\omega')\big|_{\mathbf{Y}=\mathbf{Y}^{\mathrm{Ito}}(\omega'')}.$$

defines measurable processes (in the sense they measurable with respect to $\mathcal{F}_t' \otimes \mathcal{F}_t^{Y} \;=\; \mathcal{F}_t' \otimes \mathcal{F}_t^{\mathbf{Y}^{\mathrm{It\hat{o}}}} \;\subset\; \mathcal{F}_t' \otimes \mathcal{F}_t''.$).

To connect with the original stochastic filtering problem, consider the Itô SDE on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\begin{aligned} dX_t &= b_{t,\omega}(X_t)\,dt + \sigma_{t,\omega}(X_t)\,dB_t + f_{t,\omega}(X_t)\,dY_t, \qquad X_0 = X_0(\omega'); \\ dZ_t &= Z_t h_{t,\omega}(X_t)\,dY_t, \qquad Z_0 = 1, \end{aligned} \tag{9}$$

with $\phi_{t,\omega}(x) = \phi(t, x; \mathbf{Y}^{\mathrm{Ito}}_{\cdot\wedge t}(\omega''))$, $\phi := b, \sigma, f, h$. Under *Assumption CME*, standard theory ensures a unique strong solution, denoted by (X, Z).

Theorem 20

Suppose Assumption CME and Assumption OWP hold. Then there exists a continuous modification of $(\bar{X}, \bar{Z})$*, denoted also by* $(\bar{X}, \bar{Z})$*, such that* (X, Z) *and* $(\bar{X}, \bar{Z})$ *are indistinguishable on* $(\Omega, \mathcal{F}, \mathbb{P})$*. In particular,* Z_t*,* $\bar{Z}_t$ *are integrable for any* $t \in [0, T]$*. Moreover, we have:*

(1) *for any bounded measurable function* φ*, a.s.*

$$\mathbb{E}\left[\varphi(X_t) Z_t \,\middle|\, \mathcal{F}_t^Y\right](\omega) = \mu_t^{\mathbf{Y}}(\varphi)\Big|_{\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')};$$

(2) *for any bounded measurable function* φ*, a.s.*

$$\frac{\mathbb{E}\left[\varphi(X_t)\, Z_t \,\middle|\, \mathcal{F}_t^Y\right](\omega)}{\mathbb{E}\left[Z_t \,\middle|\, \mathcal{F}_t^Y\right](\omega)} = \varsigma_t^{\mathbf{Y}}(\varphi)\Big|_{\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')}.$$

Proof of Theorem 20

According to Theorem 3.9 in [FLZ25], $(\bar{X}, \bar{Z})$ admits a continuous modification such that (X, Z) and $(\bar{X}, \bar{Z})$ are indistinguishable.
Moreover, Assumption CME and Assumption OWP imply $\mathbb{E}[\sup_{t\in[0,t]} |Z_t|^p] < \infty, \forall p \geq 1$ (which also implies the integrability of $\bar{Z}$).

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Moreover, Assumption CME and Assumption OWP imply $\mathbb{E}[\sup_{t\in[0,t]} |Z_t|^p] < \infty$, $\forall p \geq 1$ (which also implies the integrability of $\bar{Z}$).

We just need to prove (1), since it immediately implies (2). By definition and the remark on the measurability of $(\bar{X}_t, \bar{Y}_t)$, we have that

$$\mathrm{Law}\Big((\bar{X}_t, \bar{Z}_t) \mid \mathcal{F}_t^Y\Big) = \mathrm{Law}\Big((X_t^{\mathbf{Y}}, Z_t^{\mathbf{Y}})_{|\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')}\Big),$$

which means that for any bounded measurable function $F : \mathbb{R}^{d_X} \times \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}\Big[F(\bar{X}_t, \bar{Z}_t) \Big| \mathcal{F}_t^Y\Big](\omega) = \mathbb{E}\Big[F(X_t^{\mathbf{Y}}, Z_t^{\mathbf{Y}})\Big]_{|\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')}$$

Then by a standard cutoff argument and the dominated convergence theorem, for any measurable function F with $\mathbb{E}\big[|F(\bar{X}_t, \bar{Z}_t)|\big] < \infty$, the above identity also holds, which implies (1) choosing $F(x, z) = \varphi(x)z$. $\square$

Connection rough and stochastic filtering

Stochastic vs rough filtering equations

Theorem 21

Suppose Assumption CME *and Assumption* OWP *hold. Let* $\mu^{\mathbf{Y}} = (\mu^{\mathbf{Y}}_t)_{t\in[0,T]}$ *be the solution of the rough Zakai equation*

$$\mu^{\mathbf{Y}}_t(\varphi) = \mu^{\mathbf{Y}}_0(\varphi) + \int_0^t \mu^{\mathbf{Y}}_r\left(A^{\mathbf{Y}}_r\varphi\right)\, dr + \int_0^t \mu^{\mathbf{Y}}_r\left(\Gamma^{\mathbf{Y}}_r\varphi\right)\, d\mathbf{Y}_r,$$

for all test functions $\varphi \in {C_b}^3(\mathbb{R}^{d_X})$*.*
Let

$$\mu_t(\omega)(\varphi) := \mathbb{E}[\varphi(X_t)Z_t|\mathcal{F}^Y_t](\omega) \overset{\text{Th. (20)}}{=} \mu^{\mathbf{Y}}_t(\varphi)|_{\mathbf{Y}=\mathbf{Y}^{\mathrm{Ito}}(\omega'')}, \qquad t\in[0,T],$$

be the randomization of $\mu^{\mathbf{Y}}_t$*.*

Then $\mu = (\mu_t)_{t\in[0,T]}$ *is a solution of the classical Zakai equation*

$$\mu_t(\varphi) = \mu_0(\varphi) + \int_0^t \mu_r(A_r\varphi)\, dr + \int_0^t \mu_r(\nabla\varphi\, f(r,\cdot,Y_r) + \varphi\, h(r,\cdot,Y_r))\, dY_r,$$

for all test functions $\varphi \in C_b^3(\mathbb{R}^{d_X})$*.*

Idea: Note that for

$$\phi(t,\mathbf{Y}) := \mu_t^{\mathbf{Y}}(A_t^{\mathbf{Y}}\varphi), \quad \mu_t^{\mathbf{Y}}(\Gamma_t^{\mathbf{Y}}\varphi), \quad \mu_t^{\mathbf{Y}}(\Gamma_t^{\mathbf{Y}}\Gamma_t^{\mathbf{Y}}\varphi + (\Gamma_t^{\mathbf{Y}})'\varphi), \qquad \phi : [0,T] \times \mathcal{C}_T \to \mathbb{R},$$

the map ϕ is measurable and causal. Then according to Theorem 3.9 in [FLZ25], we have

$$\mu_t^{\mathbf{Y}}(\varphi)|_{\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')} = \mu_0^{\mathbf{Y}}(\varphi) + \int_0^t \mu_r^{\mathbf{Y}}(A_r^{\mathbf{Y}}\varphi)|_{\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')}\, dr + \int_0^t \mu_r^{\mathbf{Y}}(\Gamma_r^{\mathbf{Y}}\varphi)|_{\mathbf{Y}=\mathbf{Y}^{\text{Itô}}(\omega'')}\, dY_r,$$

Since

$$\mu_0^{\mathbf{Y}}(\varphi) = \mathbb{E}[\varphi(X_0)] = \mu_0(\phi) \quad \text{for any } \mathbf{Y} \in \mathcal{C}_T,$$

the result follows immediately from Theorem 20. $\square$

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Thank you for your kind attention!